

História da Computação

De 1980 a 1989

9. Evolução dos Computadores Pessoais sua Interconexão

- Os anos 1980 viram duas grandes mudanças na área da computação, as quais moldaram a nova sociedade tecnológica na qual vivemos. O primeiro movimento veio no início da década com a popularização do computador pessoal. Circuitos integrados cada vez mais poderosos e baratos possibilitaram a construção de computadores suficientemente potentes para fazer coisas úteis e ainda assim a um preço acessível para famílias e pequenas empresas.
- O segundo grande movimento foi no sentido da interligação desses computadores em rede. Foi nessa década que a Internet, que já consolidava várias redes de computadores ao redor do mundo começou a ser acessível aos usuários domésticos e pequenas empresas. No finalzinho da década, também, foi criada a World Wide Web, que remodelou a maneira como a sociedade humana funciona.
- Os anos 1980 também viram a ascensão meteórica e a queda ainda mais meteórica da indústria de videogames. Surgiram também nesse período o computador padrão PC, os processadores RISC, os notebooks e até o computador de bolso.
- Em termos de software, várias linguagens como Ada, C++, Turbo Pascal, Eiffel e Perl surgiram, além de novos sistemas operacionais como o MS-DOS, o Mac OS e o GNU. Foram anos de grandes e incríveis avanços, como veremos.

9.1 D-BASE II – 1980



Wayne Ratliff (Estados Unidos, 1946)

```
www.hpmuseum.net
B>do

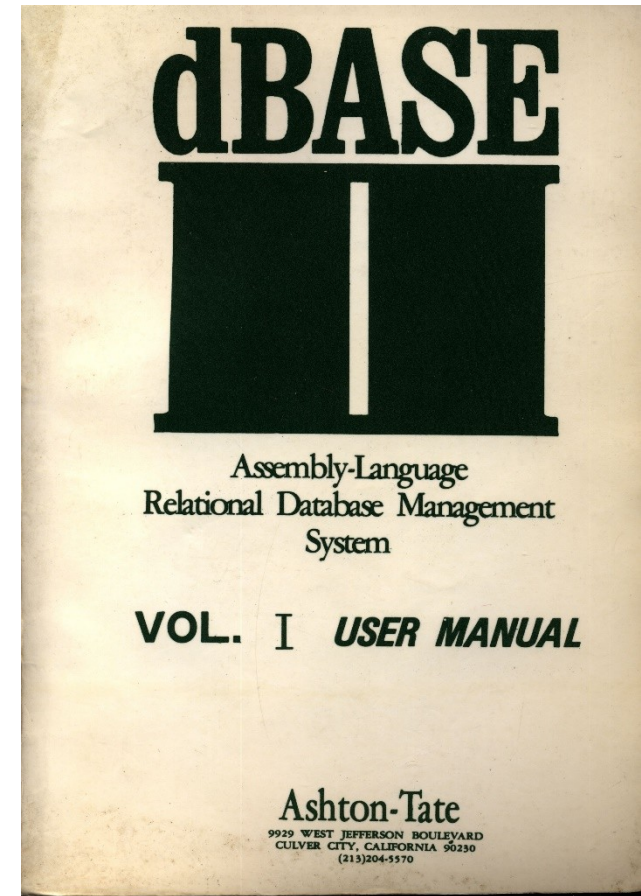
*** dBASE II/86 Ver 2.41 6 JUNE 1984

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*  HELP  LIST  LIST  MODI  48  3  MODI  BROWSE  EDIT #  QUIT
   FILES STRUC STRUC                                     COMM
   Num Pad 15:10
```



9.2 Linguagem Ada – 1980

Ada Program Structure

```
package Point_Pkg is
  procedure Assign(X, Y : in Float);
  -- Assigns X and Y as the new coordinates

  procedure Fetch(X, Y : out Float);
  -- Copies the current coordinates to X and Y
end Point_Pkg;
```

```
package body Point_Pkg is

  X, Y : Float; -- Coordinates

  procedure Assign(X, Y : in Float) is
  begin
    Point_Pkg.X := X;
    Point_Pkg.Y := Y;
  end Assign;

  procedure Fetch(X, Y : out Float) is
  begin
    X := Point_Pkg.X;
    Y := Point_Pkg.Y;
  end Fetch;
end Point_Pkg;
```

```
with Point_Pkg;
procedure Prog is
  X, Y : Float;
begin
  Point_Pkg.Assign(1.0, 2.0);
  Point_Pkg.Fetch(X, Y);
end Prog;
```

Figure 1

Shown here is an example of Ada program structure.



9.3 Commodore VIC-20 – 1980





WHY BUY JUST A VIDEO GAME?

“FOR UNDER \$300 GET THE COMMODORE VIC-20,”
THE WONDER COMPUTER OF THE 1980s.”
—William Shatner

“An investment that grows with your family needs.”



For about the cost of a video game, you can own the Commodore VIC-20™, a full-fledged color computer that's so easy to use even a child can be computing in minutes.

Sure it plays the great games kids love. But the VIC-20 can also improve learning skills. In fact, it uses the same computer language taught in school on the Commodore PET™. So students learning on the PET in class, can practice computing at home on the VIC.

Unlike video game machines, the VIC-20 has many practical applications, such as financial planning and word processing. And the VIC-20 is never out of date. Because, unlike video machines, it's completely expandable. With low priced accessories, including a tape recorder and disk drive, to store and retrieve information, printer and our exciting new VIC MODEM™ — which lets you communicate with outside information sources.

The wonder computer of the 1980s — VIC-20 from Commodore. Under \$300. At your Commodore dealer and selected stores.



VIC-20 VS. THE VIDEO GAMES

Product Features	VIC-20	Atari VCS®	Mattel Intellivision®
Under \$300	YES	YES	YES
Plays Cartridge Games	YES	YES	YES
A Real Computer	YES	NO	NO
Full-Size Computer Keyboard	YES	NO	NO
Basic Computing Language Built-In	YES	NO	NO
Expandable Memory Capability	YES	NO	NO
Self-Teaching Programming Manual	YES	NO	NO
Works With Printer	YES	NO	NO
Also Works With Disks and Cassettes	YES	NO	NO



Commodore Computer Systems
601 Moore Rd., King of Prussia, PA 19406
Canadian Residents: Commodore Computer Systems
3070 Pharmacy Avenue, Agincourt, Ontario, Canada, M1W 2K4
Please send me more information on the VIC-20.

Name _____
Address _____
City _____ State _____ Zip _____
Phone _____

9.4 Apple III – 1980



9.5 Sinclair ZX80 – 1980



Sir Clive Sinclair
(Reino Unido, 1940)

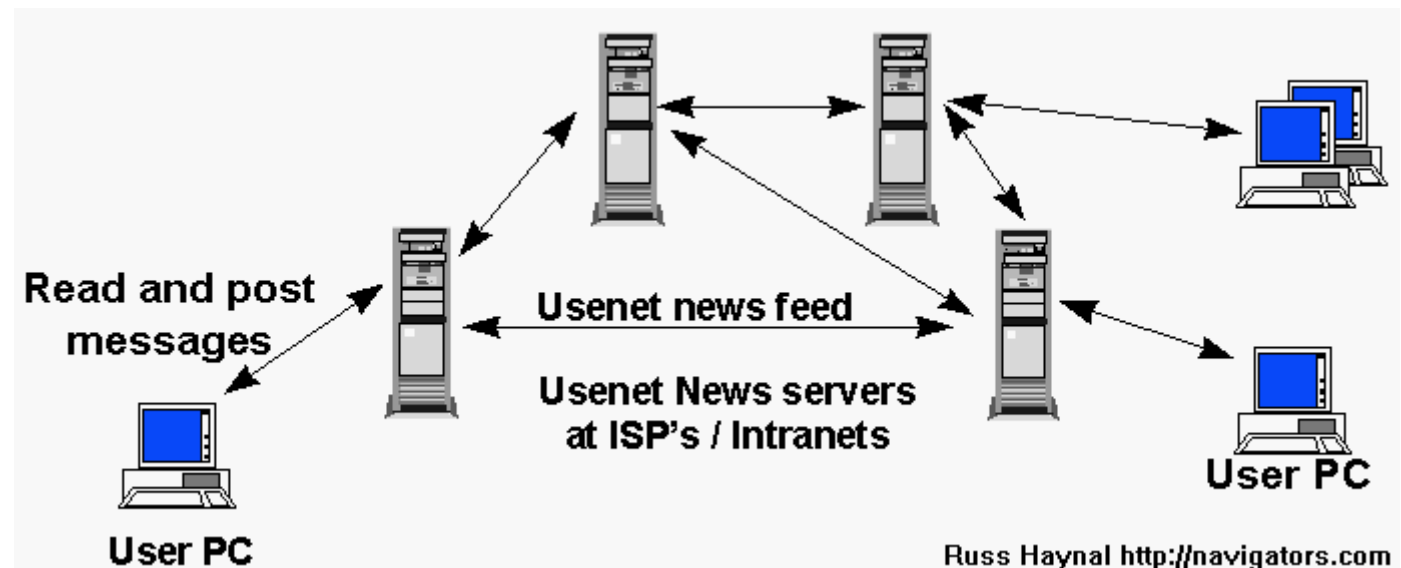
9.6 UseNet – 1980



Tom Truskott



Jim Ellis (Estados Unidos, 1956-2001)



9.7 MS-DOS – 1980



```
Current date is Tue 1-01-1980
Enter new date:
Current time is 21:35:24.18
Enter new time:

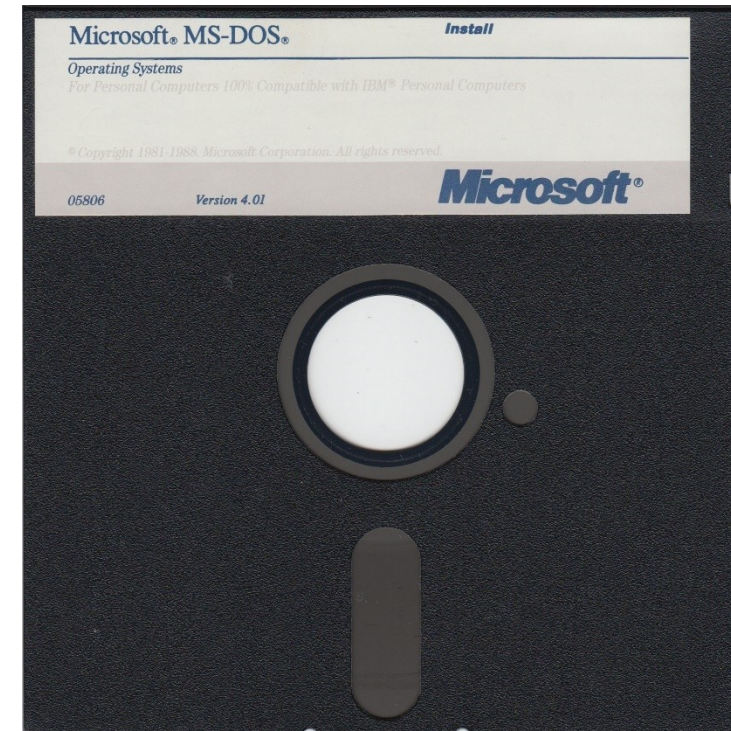
The IBM Personal Computer DOS
Version 2.00 (C)Copyright IBM Corp 1981, 1982, 1983

A>dir

Volume in drive A has no label
Directory of A:\

COMMAND  COM      17664   3-08-83  12:00p
FORMAT   COM       6016   3-08-83  12:00p
CHKDSK    COM       6400   3-08-83  12:00p
SYS       COM       1408   3-08-83  12:00p
DEBUG     COM      11904   3-08-83  12:00p
SLOOP          32     1-01-80   7:44p
        6 File(s)  292864 bytes free

A>_
```



9.8 Xerox 8010 Star – 1981



XEROX 6085 Workstation
User-Interface Design

To make it easy to compose text and graphics, to do electronic filing, printing, and mailing all at the same workstation, requires a revolutionary user interface design.

Bit-map display - Each of the pixels on the 19" screen is mapped to a bit in memory; thus, arbitrarily complex images can be displayed. The 6085 displays all fonts and graphics as they will be printed. In addition, familiar office objects such as documents, folders, file drawers and in-baskets are portrayed as recognizable images.

The mouse - A unique pointing device that allows the user to quickly select any text, graphic or office object on the display.

See and Point

All functions are visible to the user on the keyboard or on the screen. The user does filing and retrieval by selecting them with the mouse and touching the MOVE, COPY, DELETE or PROPERTIES command keys. Text and graphics are edited with the **TABLE** keys.

Shorter Production Times

Experience at Xerox with prototype workstations has shown shorter production times and thus lower costs, as a function of the percentage of use of the workstations. The following equation can be used to express this:

Activity under the old and the new

Figure 7: Data from Table 1 drive

Workstation usage percentages Table 1 and illustrated in Figure 6085 users are likely to do less composition and layout, extra process including printing and d

Text and Graphics

To replace typesetting, the 6085 offers a choice of type fonts and sizes from 6 point to 36 point:

Here is a sentence of 18-point text.
18-point text.
24-point text.
36-point text.

12294 Free Disk Pages

9:27:24 10-29-80

Local Kevin J. Outbasko

Mail Merge Mail from Ken

Calendar Calc Loader

Blank User Dictionary Empty Dictionary Blank Record File Blank Document

2.0 TTY Monthly Profit Blank Folder

Beethoven

C Tools

PC

Emulator

Emulated Rigid Disk

Virtual Floppy

Example ViewPo

Remote Files

1427

DOS & Lotus

Blank Reference

Wastebasket

Directory

Drawers in Japan

Mickey

OS/2

Xerox

Tape Drive

Floppy Drive

Wastebasket

Directory

9.9 IBM PC – 1981



William C. Lowe (Estados Unidos, 1941-2013)



9.10 Placa de Video CGA – 1981



9.11 Revista Micro Sistemas – 1981



Alda Campos

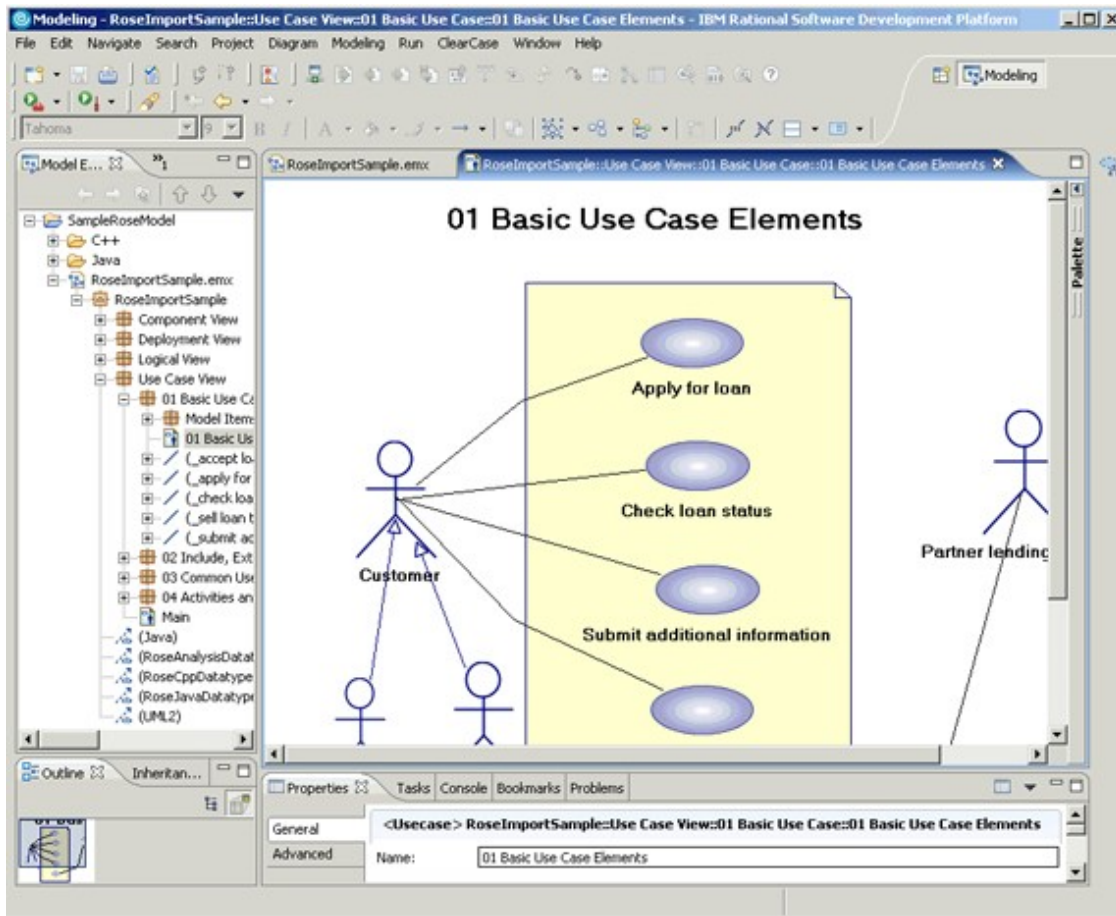


9.12 Osborne 1 – 1981



Adam Osborne (Tailândia, 1939-2003)

9.13 Rational Machines – 1981



The three amigos



Grady Booch



James Rumbaugh



Ivar Jacobson

9.14 Sun Microsystems Inc. – 1982



Sun-1

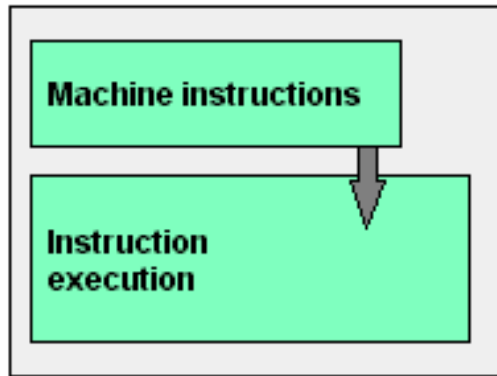


9.15 Commodore 64 – 1982

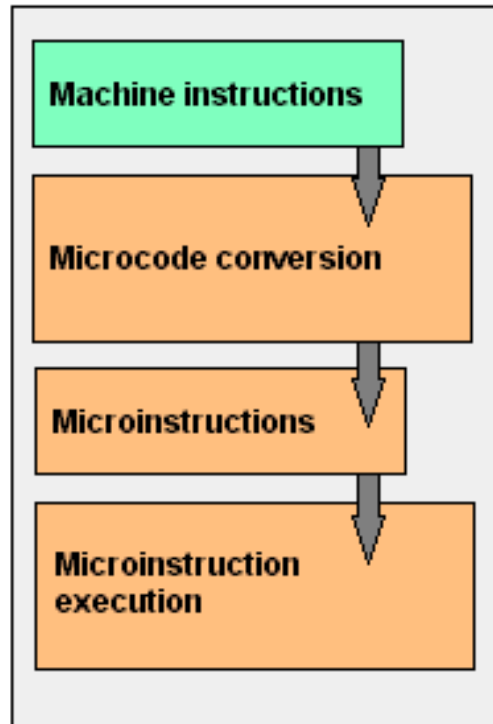


9.16 Arquitetura RISC – 1982

RISC



CISC



Carlo H. Séquin (Suíça, 1941)



David Patterson (Estados Unidos, 1947)

9.17 BBC Micro – 1982



Front Row: David Kitson, Chris Searle

2nd Row: Chris Turner, David Allen

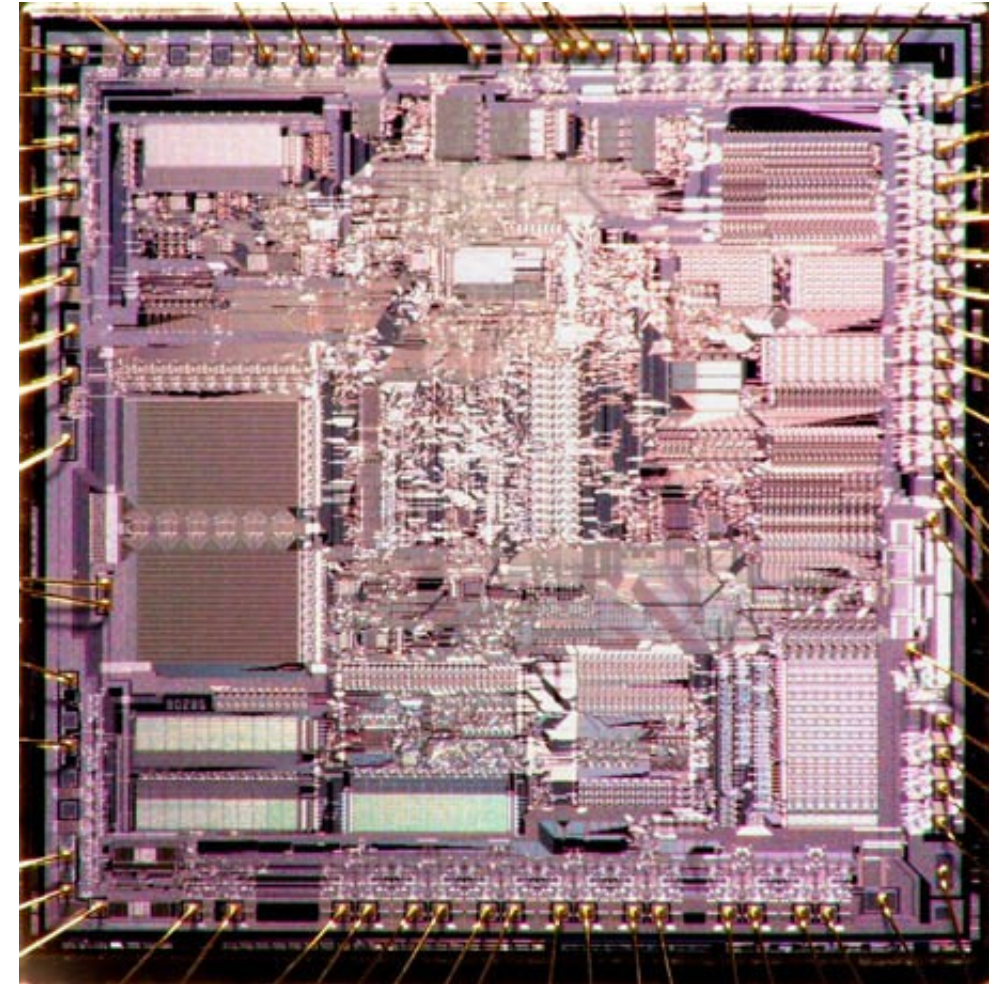
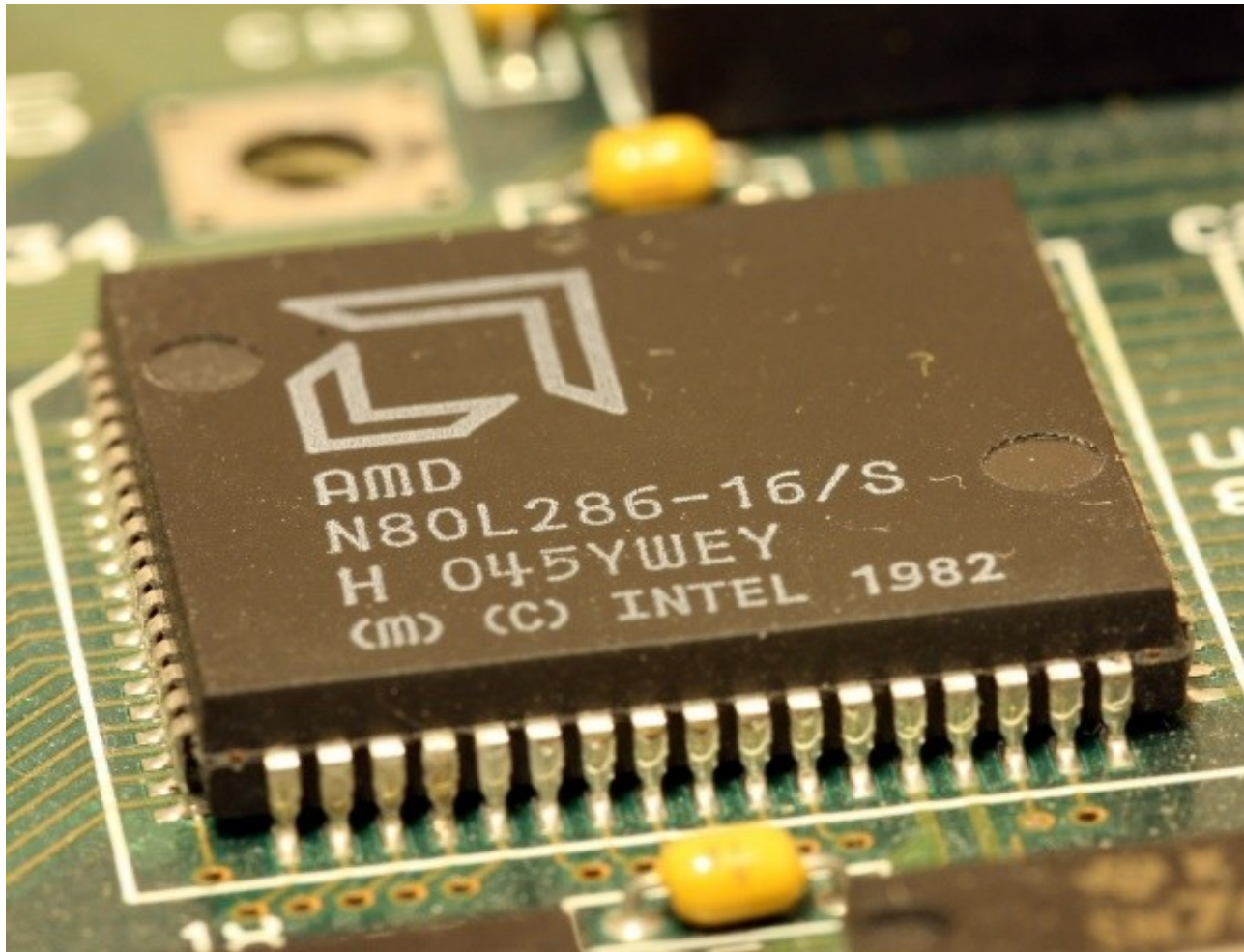
3rd Row: Steve Furber, Sophie Wilson

Back Row: Hermann Hauser, Andy Hopper, Chris Curry

9.18 GRiD Compass 1101, o Primeiro Notebook – 1982



9.19 80286 – 1982



9.20 Compac Portable – 1982



9.21 ZX Spectrum – 1982



9.22 Computadores de Quinta Geração – 1982

FIFTH GENERATION OF COMPUTER

“A bold 10-year effort by Japan to seize the lead in computer technology is fizzling to a close, having failed to meet many of its ambitious goals or to produce technology that Japan's computer industry wanted. After spending \$400 million on its widely heralded Fifth Generation computer project, the Japanese Government said this week that it was willing to give away the software developed by the project to anyone who wanted it, even foreigners.”

Source: [New York Times](#)

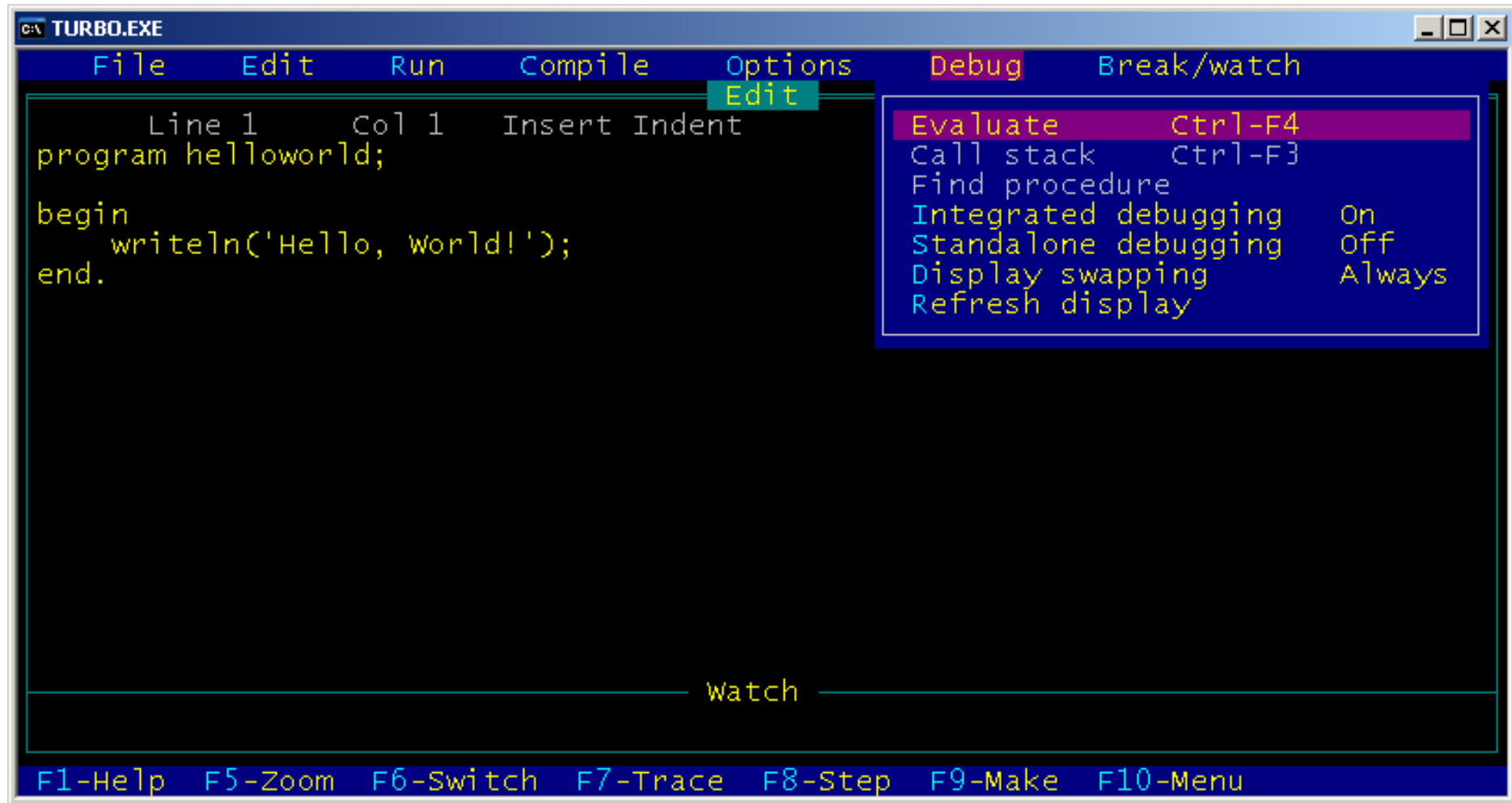
9.23 C++ – 1983

```
1  #include<iostream>
2  using namespace std;
3
4  int main() {
5      int number, reverse = 0;
6      cout<<"Input a Number to Reverse: ";
7      cin>> number;
8
9      for( ; number!= 0 ; )
10     {
11         reverse = reverse * 10;
12         reverse = reverse + number%10;
13         number = number/10;
14     }
15     cout<<"New Reversed Number is: "<<reverse;
16
17     return 0;
18 }
```

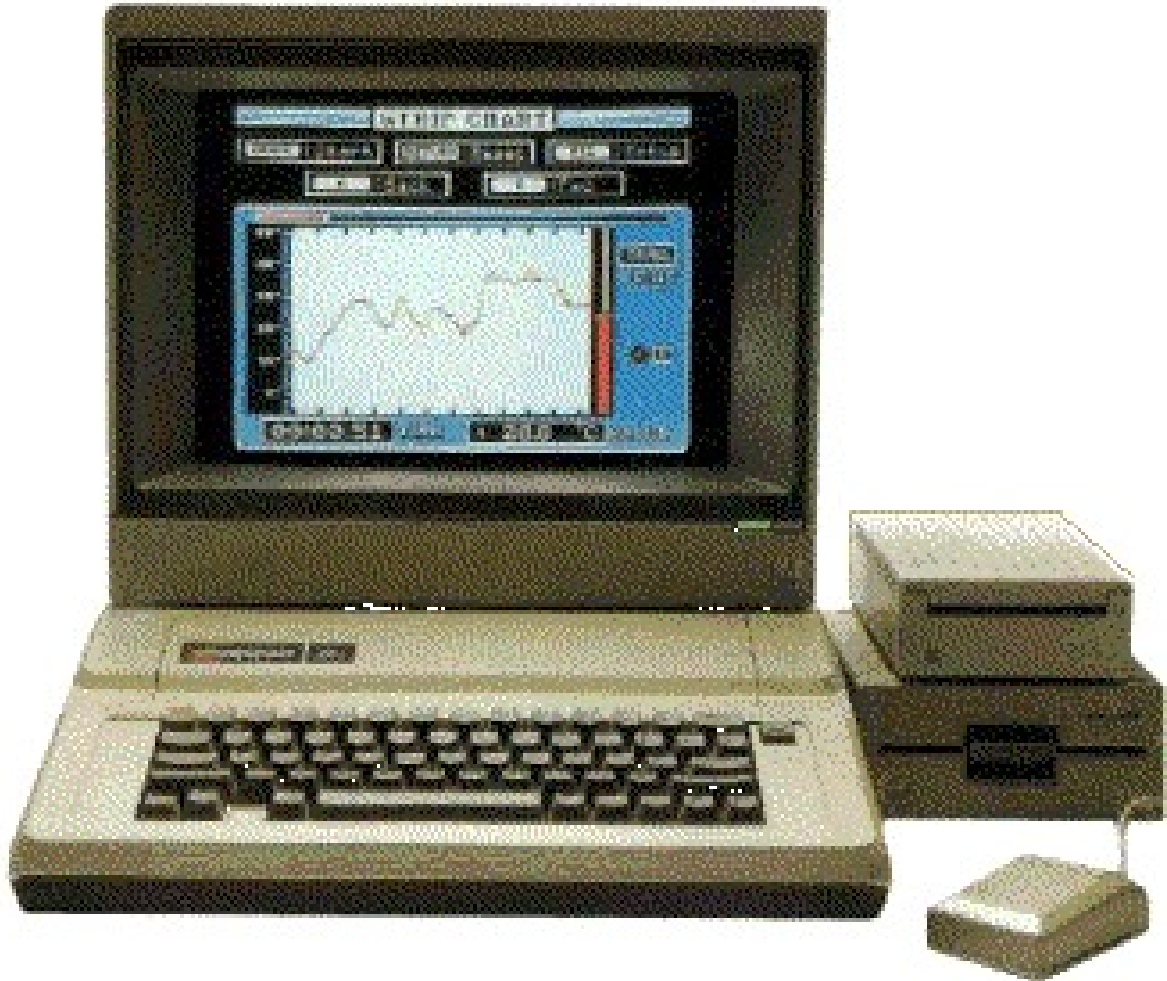


Bjarne Stroustrup (Dinamarca, 1950)

9.24 Turbo Pascal – 1983



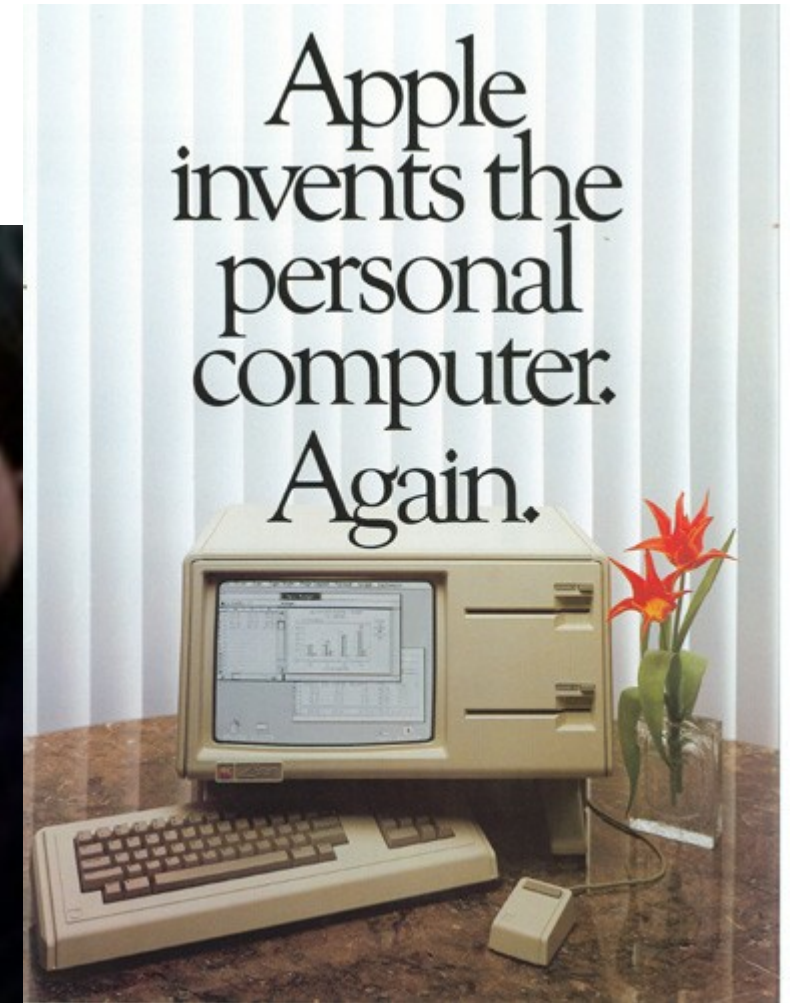
9.25 Apple IIe – 1983



9.26 Apple Lisa – 1983



Steve Jobs



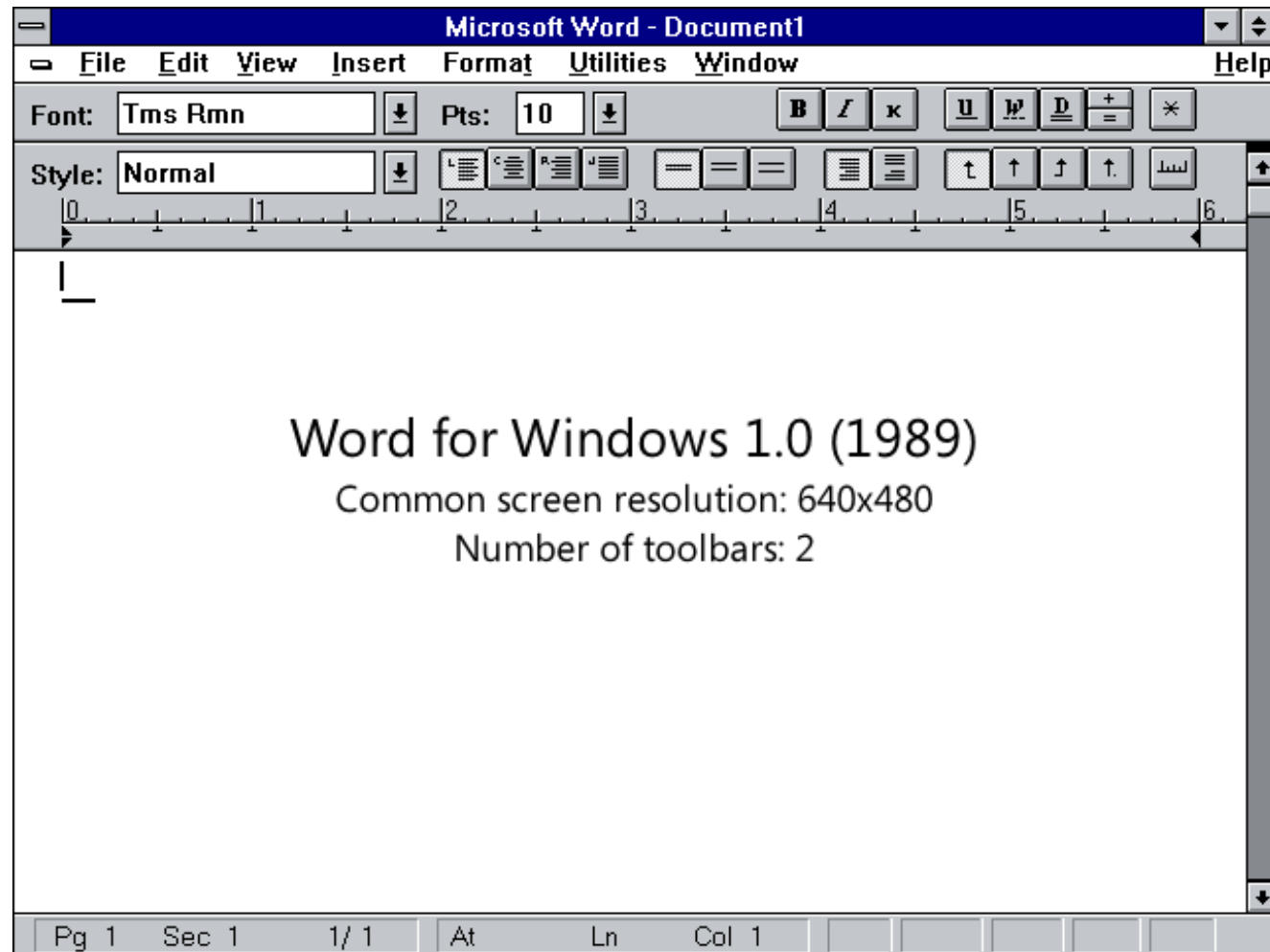
9.27 IBM PC-XT – 1983



9.28 IBM PCjr – 198

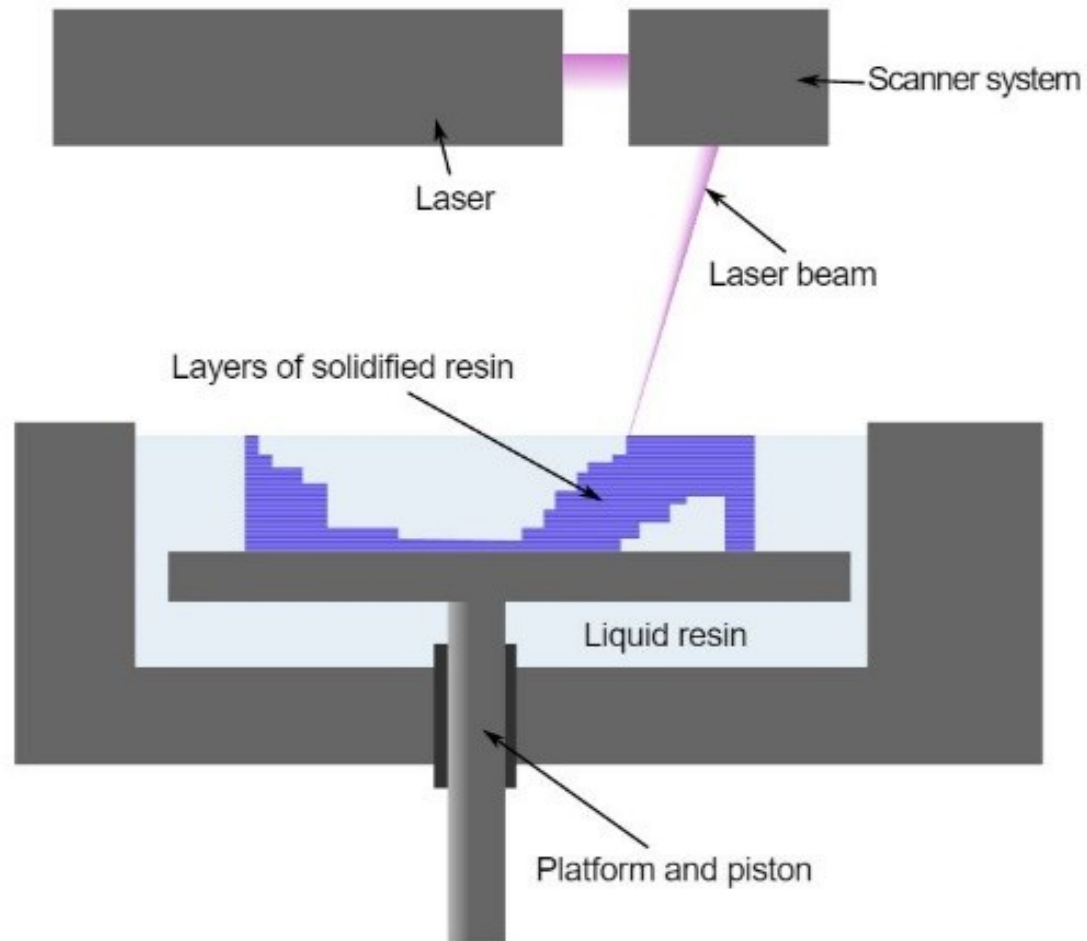


9.29 Word – 1983



Charles Simonyi (Hungria, 1948)

9.30 Impressora 3D – 1984



Charles Hull (Estados Unidos, 1939)

9.31 GNU – 1984

Richard M. Stallmann (Estados Unidos, 1953)



Free Software Free Society

A Public Lecture Followed by Q & A

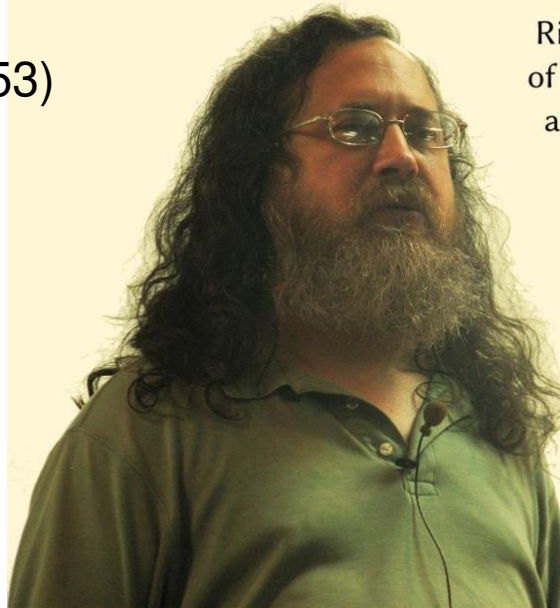
Richard Stallman

Pioneer of Free Software and Copyleft

1700-1900 hrs on January 21, 2014

Old Conference Hall, Main campus
Tata Institute of Social Sciences, Deonar,
Mumbai 400088

ALL ARE WELCOME!



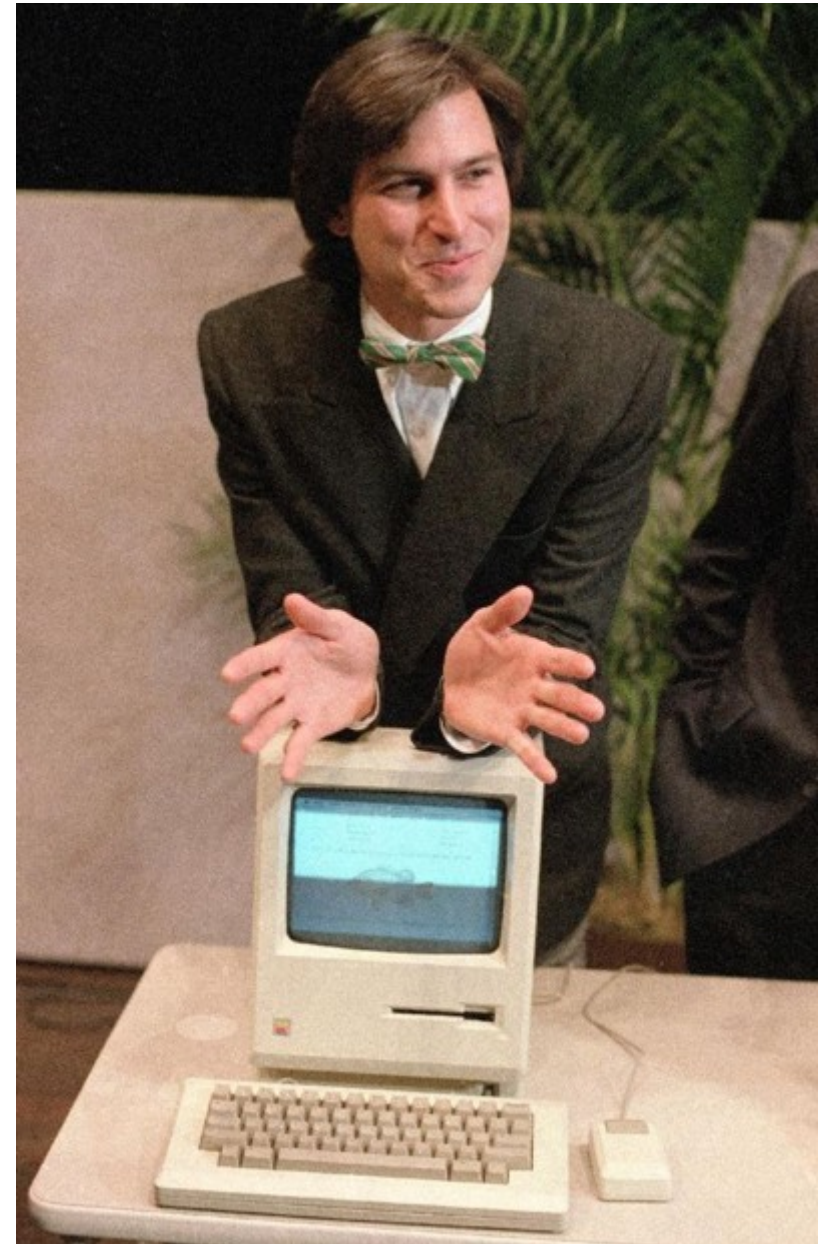
Richard Stallman is the founder of the Free Software Foundation and the GNU Project. Stallman pioneered the concept of copyleft, and is the author of the GNU General Public License (GPL), the most widely used copyleft license, for free software. He will also talk about the relevance of software freedom in the era of networked computers.

gnu.org | fsf.org | gnu.org.in

Organised by:
Tata Institute of Social Sciences,
Deonar, Mumbai 400088

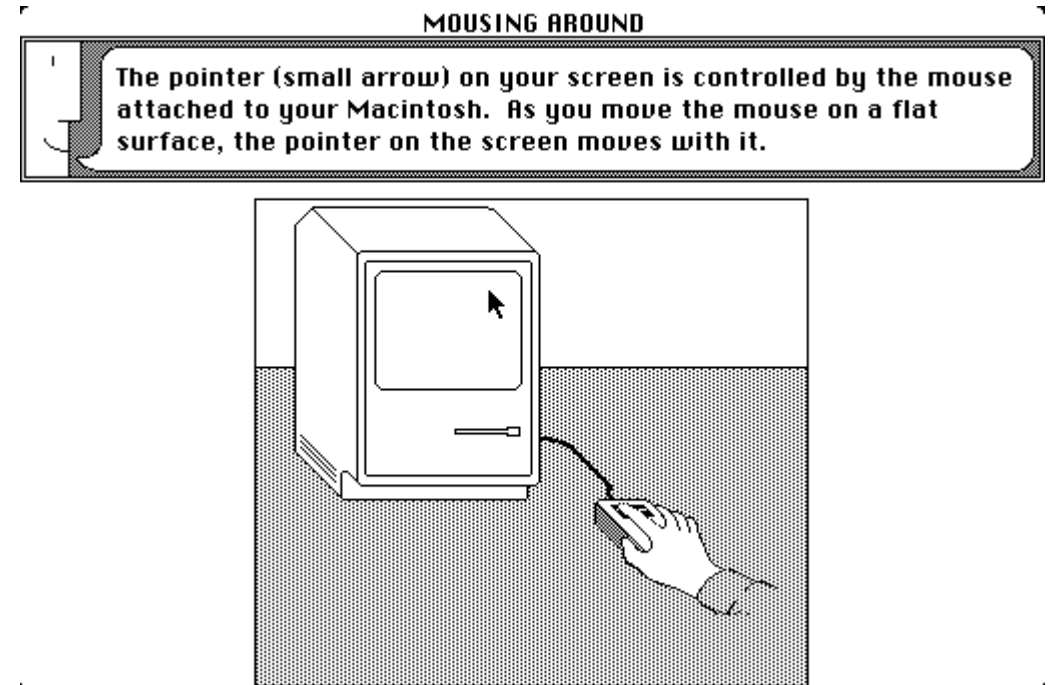
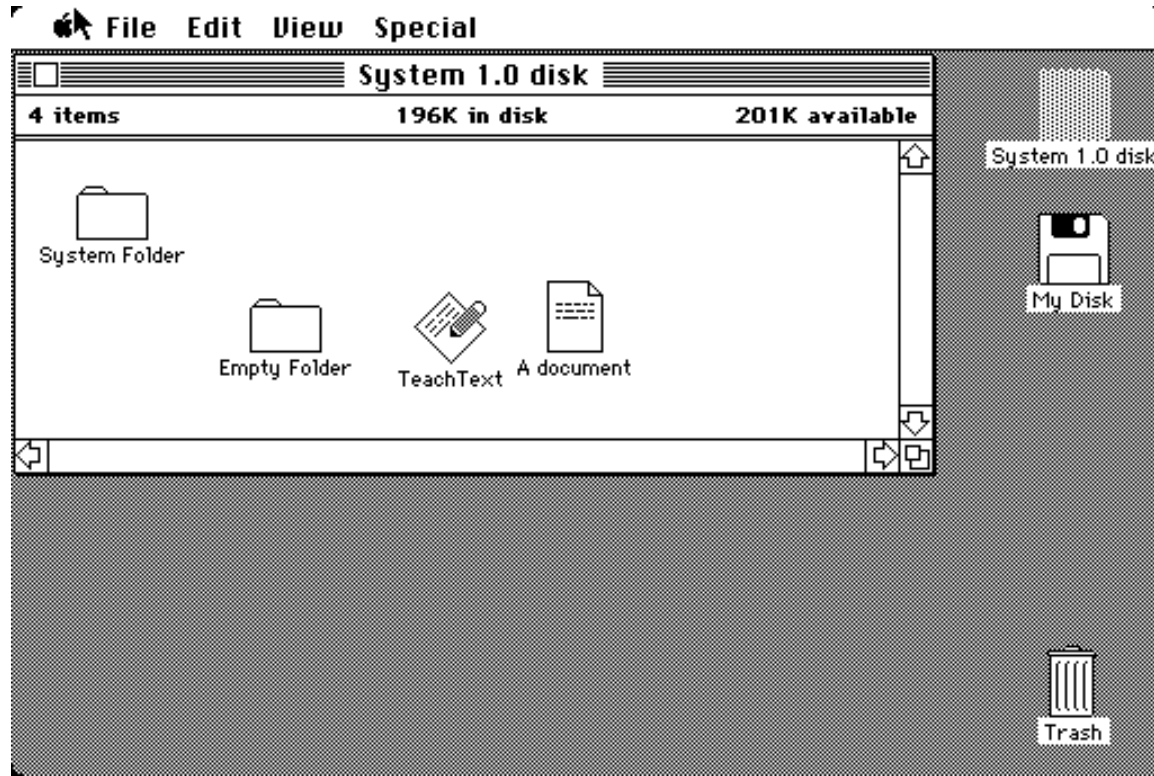
Contact:
Aravindhan: 9892515519
and gnu@gnu.org.in

9.32 Apple Macintosh – 1984



<https://www.youtube.com/watch?v=VtvjbmoDx-I>

9.33 Mac OS – 1984



9.34 IBM AT – 1984



9.35 Tetris – 1984

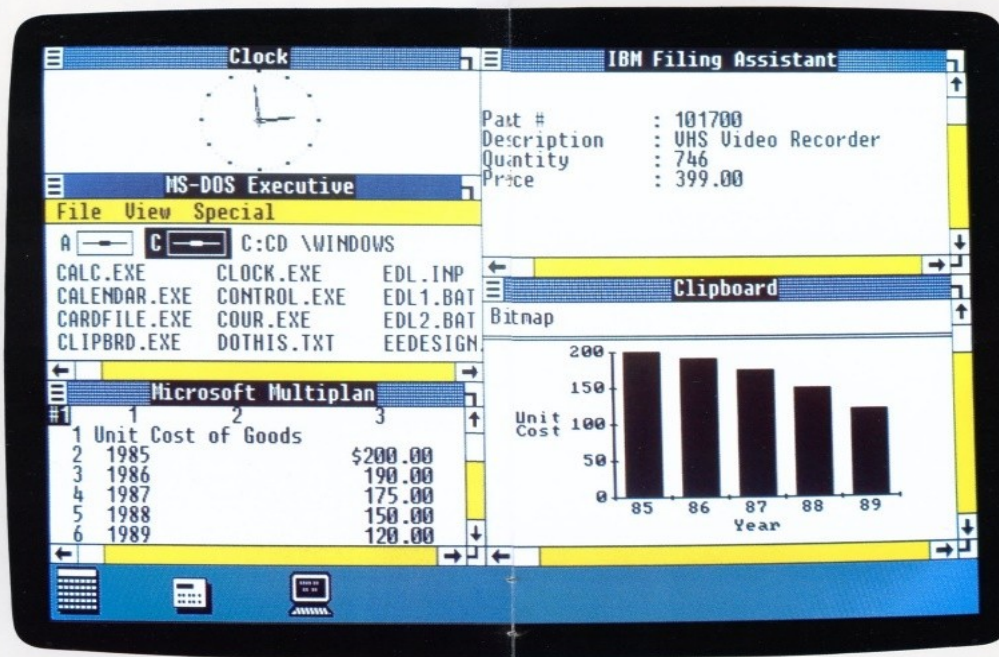


Alexey Pajitnov (Rússia, 1956)



9.36 Windows – 1985

It's going to open your eyes.



Microsoft® Windows has arrived.

For anyone who uses a computer in earnest, that is extremely good news.

Windows gives you a practical way to integrate programs. It radically decreases the time it takes to move from one application to another. Dramatically simplifies the means of consolidating data from many different programs.

And, as a graphical extension of the MS-DOS® operating system, it gives you a highly visual way to work and to organize your work.

In short, Windows brings efficiency to all those processes of personal computing which have till now been awkward, unwieldy, inconvenient.

The joys of job hopping.

With the advent of Windows, you can work with multiple applications. And switch from program to program with ease.

Start up with one application, then another, and another. Leap back and forth between applications as your work routine dictates. Then pick up right where you left off.

The ability of Windows to change quickly from program to program logically and naturally magnifies the utility and productivity of the personal computer. And is a

recognition of the way people who exploit the power of PCs really do their jobs.

Breaking the 640K barrier.

Just like you, Microsoft Windows can handle several projects at the same time. Juggle assignments. Deal with frequent interruptions.

And Windows will ignore the 640K limit of your PC, especially if you have a hard disk, the Intel® Above® Board, or expanded memory. It will execute the rather neat trick of working with more programs than memory can hold at one time.

Spreading knowledge.

Another great service Windows performs is accelerating the movement of information from one program to another.

Collecting and combining that information is as simple as taking a "snapshot" of data in one program. Editing it. Then consolidating it with data from other programs.

With Windows, you can enjoy the advantages of conventional integrated programs without their compromises. Because Windows lets you put together the applications that you know, and that get a job

done for you.

Choose your best word processor, spreadsheet, database—you name it. They're all there for you at a keystroke.

Common ground.

Finally, Windows is not only an immensely powerful tool for today, it's also a solid base for a new generation of Windows applications.

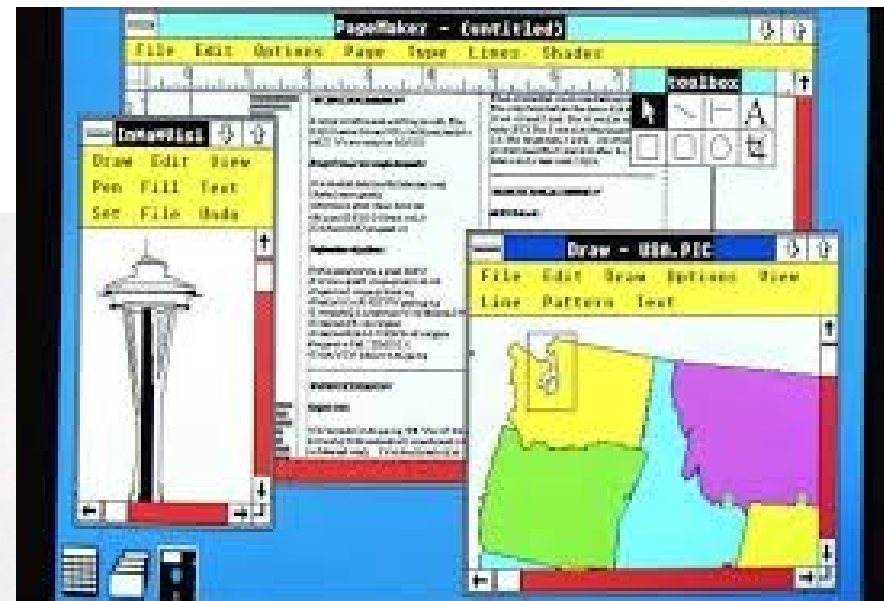
As an introductory offer, two of these—Microsoft Windows Write and Paint—are included in the package. Along with more than a dozen other programs.

In Windows applications you have a common interface which includes drop-down menus, dialog boxes, icons. Along

with a richer environment that allows you to mix pictures and text. And to summon different type faces and styles at a keystroke.

Windows is a bridge between today's applications and the graphics based software now evolving. A way to work interchangeably with today's programs. And tomorrow's.

If you're someone who uses personal computing as a natural part of your work life, who capitalizes on the productive powers of sophisticated applications, look into Windows, a new vision of what a computer can do.



9.37 Atari ST – 1985



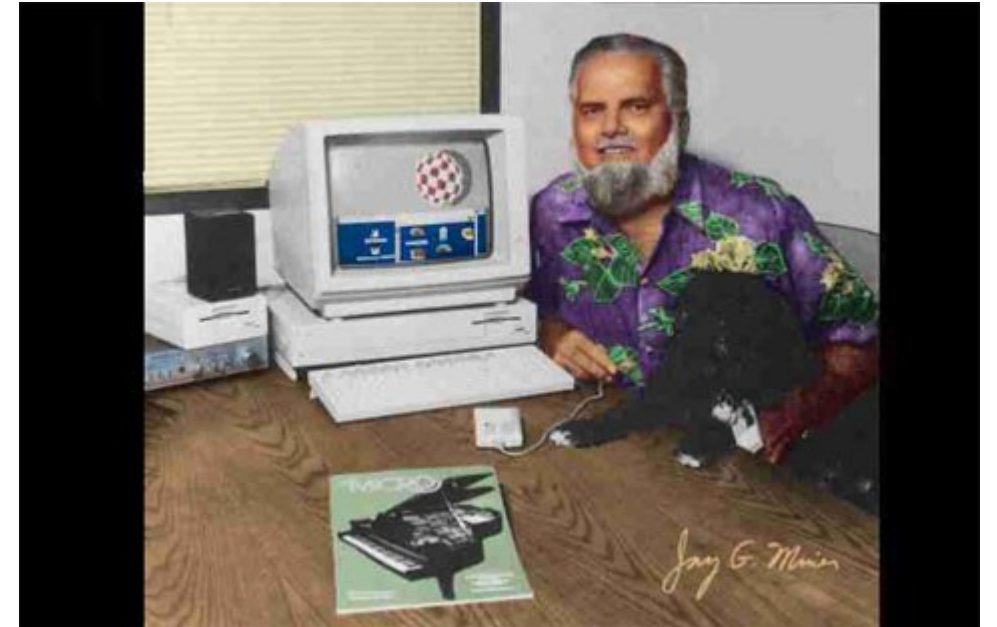
Jack Tramiel

December 13, 1928 – April 8, 2012

"We need to build computers for the masses, not the classes."

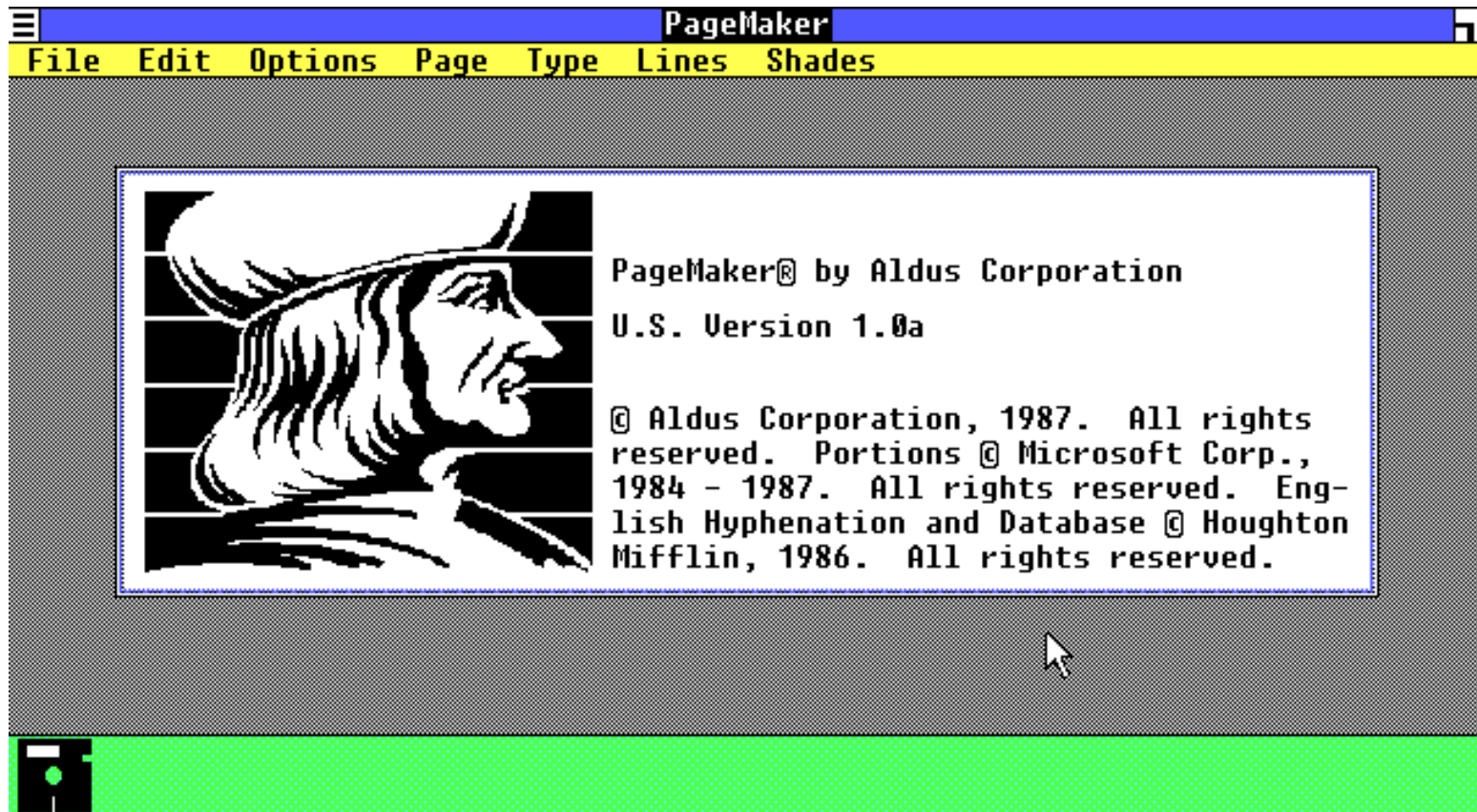


9.38 Commodore Amiga – 1985



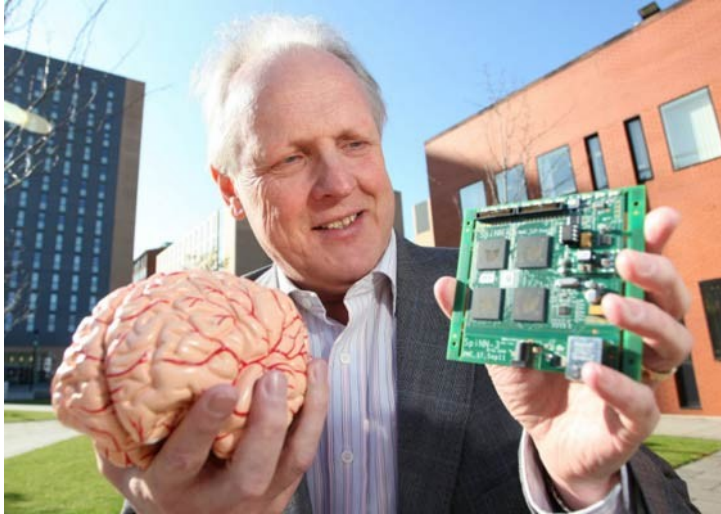
Jay Miner (Estados Unidos, 1932-1994)

9.39 PageMaker – 1985

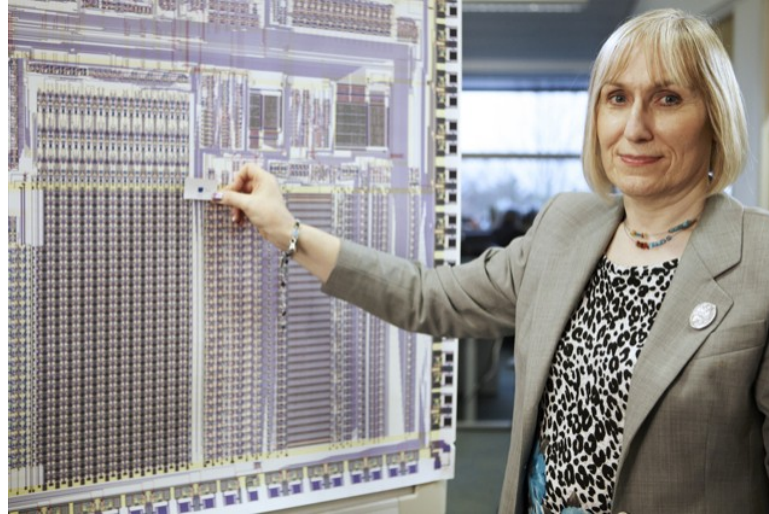


Paul Brainerd (Estados Unidos, 1947)

9.40 Acorn ARM – 1985



Steve Furber (Reino Unido, 1953)



Sophie Wilson (Reino Unido, 1957)



Roberto Heaton



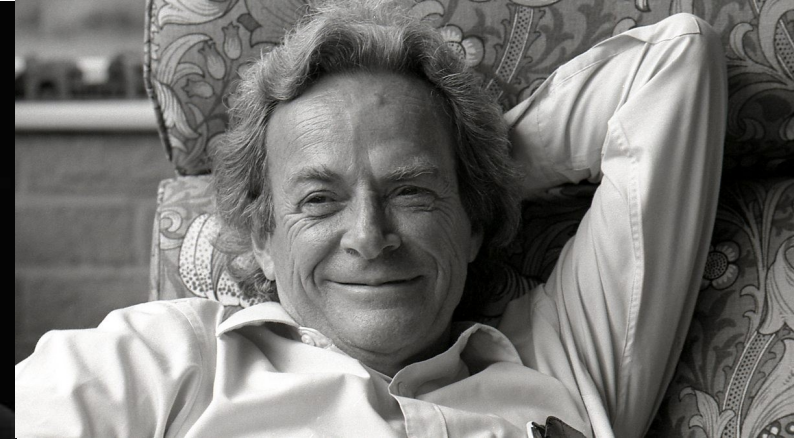
9.41 CD-ROM – 1985



9.42 Connection Machine – 1986

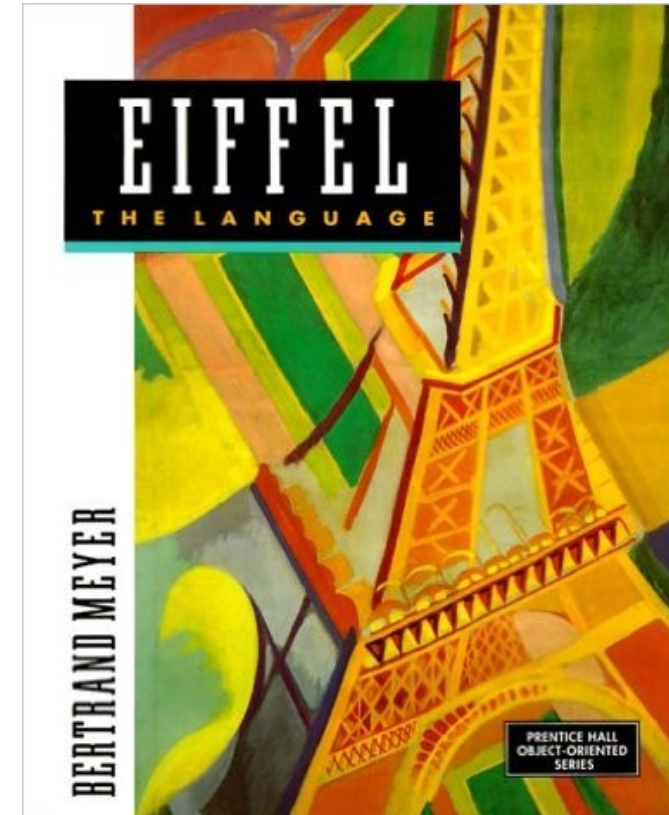
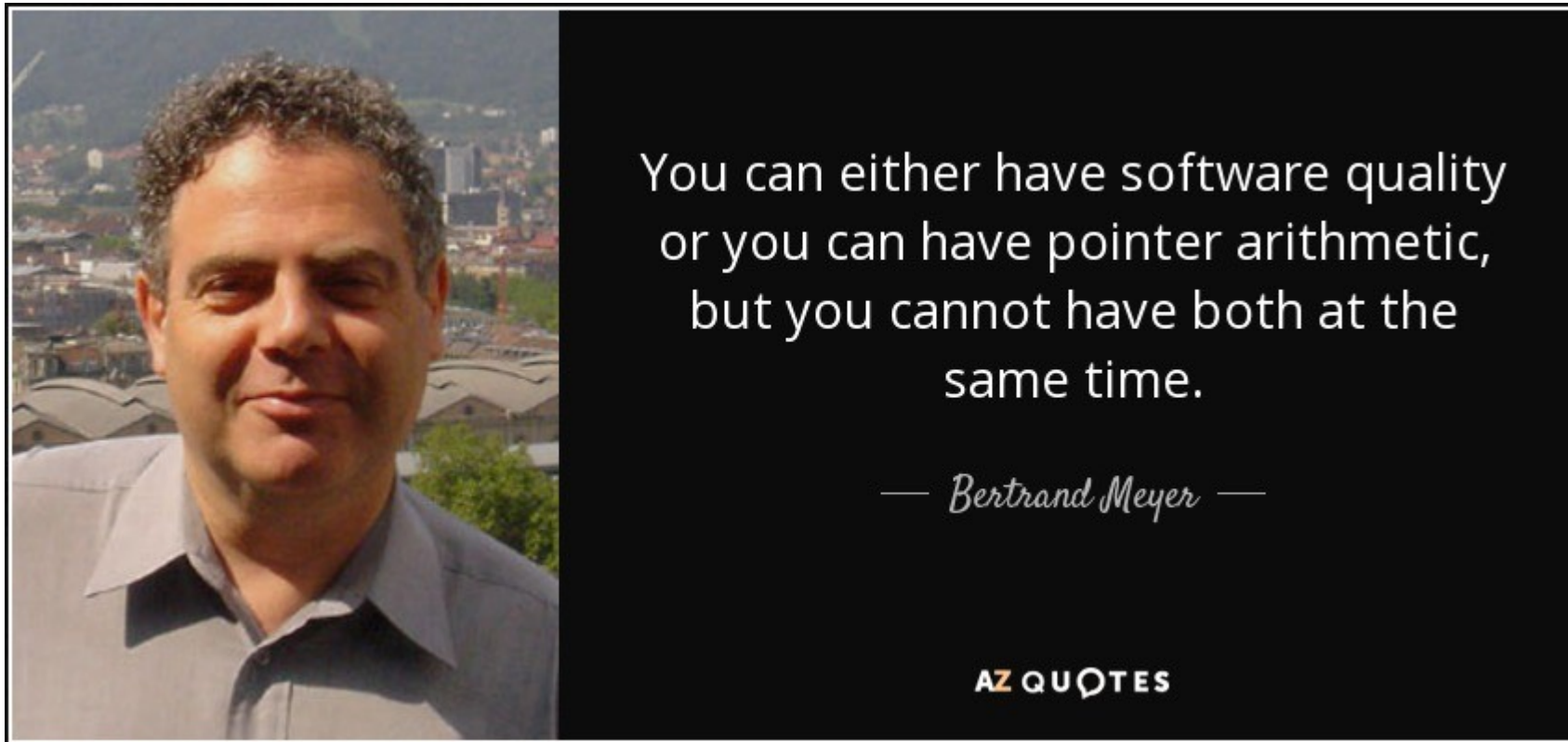


W. Daniel Hillis (Estados Unidos, 1956)



Richard Feynman (Estados Unidos, 1918-1988)

9.43 Eiffel – 1986



Bertrand Meyer (França, 1950)

9.44 Amstrad PC-1512 – 1986



Sir Alan Sugar (Reino Unido, 1947)

9.45 Compaq Deskpro 386 – 1986

COMPAQ creates the with a business



The new COMPAQ DESKPRO 386™ is the first complete high-performance PC CAD/CAM solution that single-handedly over all the popular engineering and business software. It offers versatility without compromise. Each and every component, for purposes the limits of previous "advanced technology" PCs. From its superior microprocessor to its exceptional memory capacity to its greater storage to its massive all the way to its faster disk drives, it is the most advanced personal computer in the world.

The new PC is also a true intermediate level of performance. From thousands of engineering and business programs, 3.5-inch floppy disk and hard disks.

The most advanced personal

first engineering PC background.

Microcomputer performance on your desktop

The COMPAQ DESKPRO 386 delivers microcomputer performance. Running at an impressive 10 MHz, its 80386 technology and advanced 32-bit architecture run engineering software like AutoCAD® and CHANCE® and business programs like Lotus 1-2-3® 2-3 times faster than any other advanced technology PC. Plus it welcomes boards for communicating with mainframes as well as industry-standard graphics display boards and peripherals, all critical to your design. You can also add the 5-MHz Intel 80287 coprocessor to speed graphics and floating point linear applications.

More memory to draw on

Every single component in the COMPAQ DESKPRO 386 has been optimized to take advantage of its increased speed and power. You can get up to 33 Megabytes of 32-bit high-performance RAM on the system board, 16 Megabytes overall, without waiting for new versions of DOS to use it.

More to work with

The COMPAQ DESKPRO 386 sports the latest 45, 50, and 120-Megabyte internal fixed disk drives in the industry so you can access data two times faster than other advanced technology PCs. Work the performance of the new COMPAQ Color Monitor for enhanced text and graphics resolution. Displaying 26 colors at once from a palette of 64, it comes with the COMPAQ Enhanced Color Graphics Board.

Expanded speed, enhanced graphics and the ability to run today's UNIX®-based CAD/CAM software along with thousands of industry standard business programs, make a versatile, cost-effective alternative to expensive dedicated workstations. Plus, the new COMPAQ DESKPRO 386 comes with a one-year warranty.

History in the making from a company that knows how

Long after others copy its success, the new COMPAQ DESKPRO 386 will still be the world's most advanced personal computer. Because it takes on the demands of separate subcomponents.

It's no wonder COMPAQ Personal Computers have the highest user satisfaction rating in the industry. And its members can make the PORTUNE 500 faster than any other company in history. For the Authorized Dealer nearest you, or to obtain a brochure, call 1-800-231-0900 (in Canada, call 416-445-4741) and ask for option 25.

It simply works better.

**COMPAQ
DESKPRO 386**



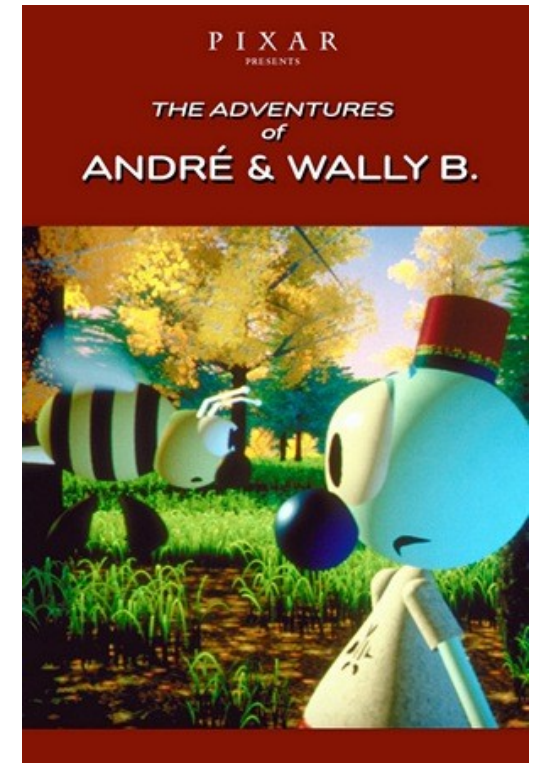
9.46 Pixar – 1986



PIXAR



ANIMATION STUDIOS



9.47 Acorn Archimedes – 1987



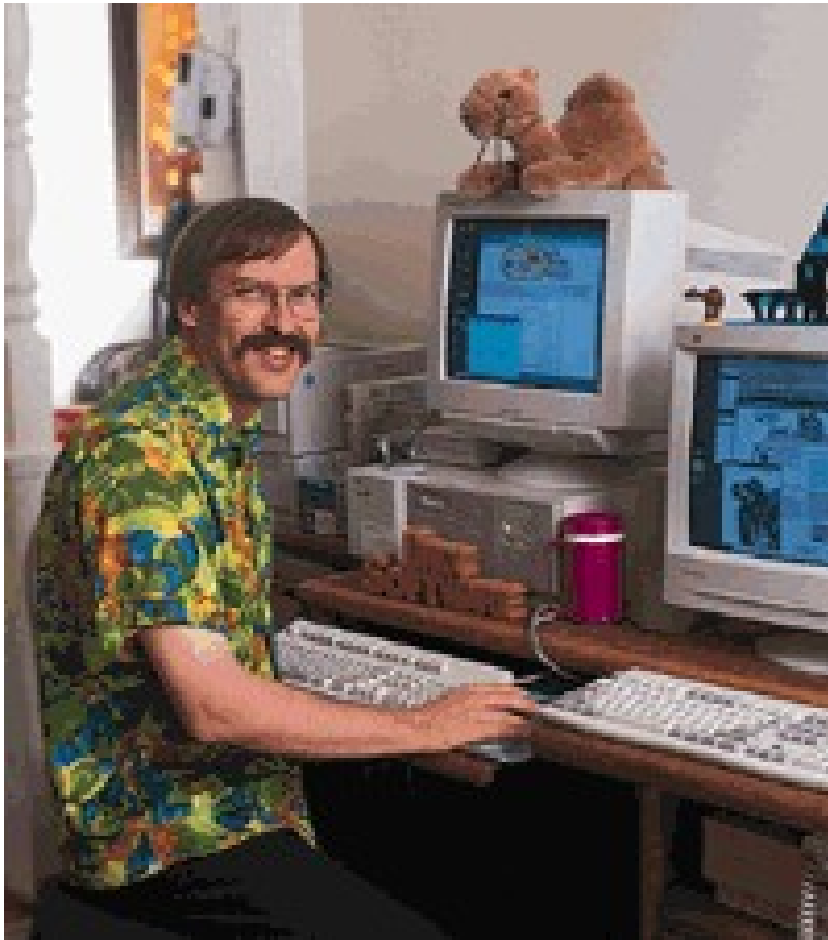
9.48 IBM PS/2 e OS/2 – 1987



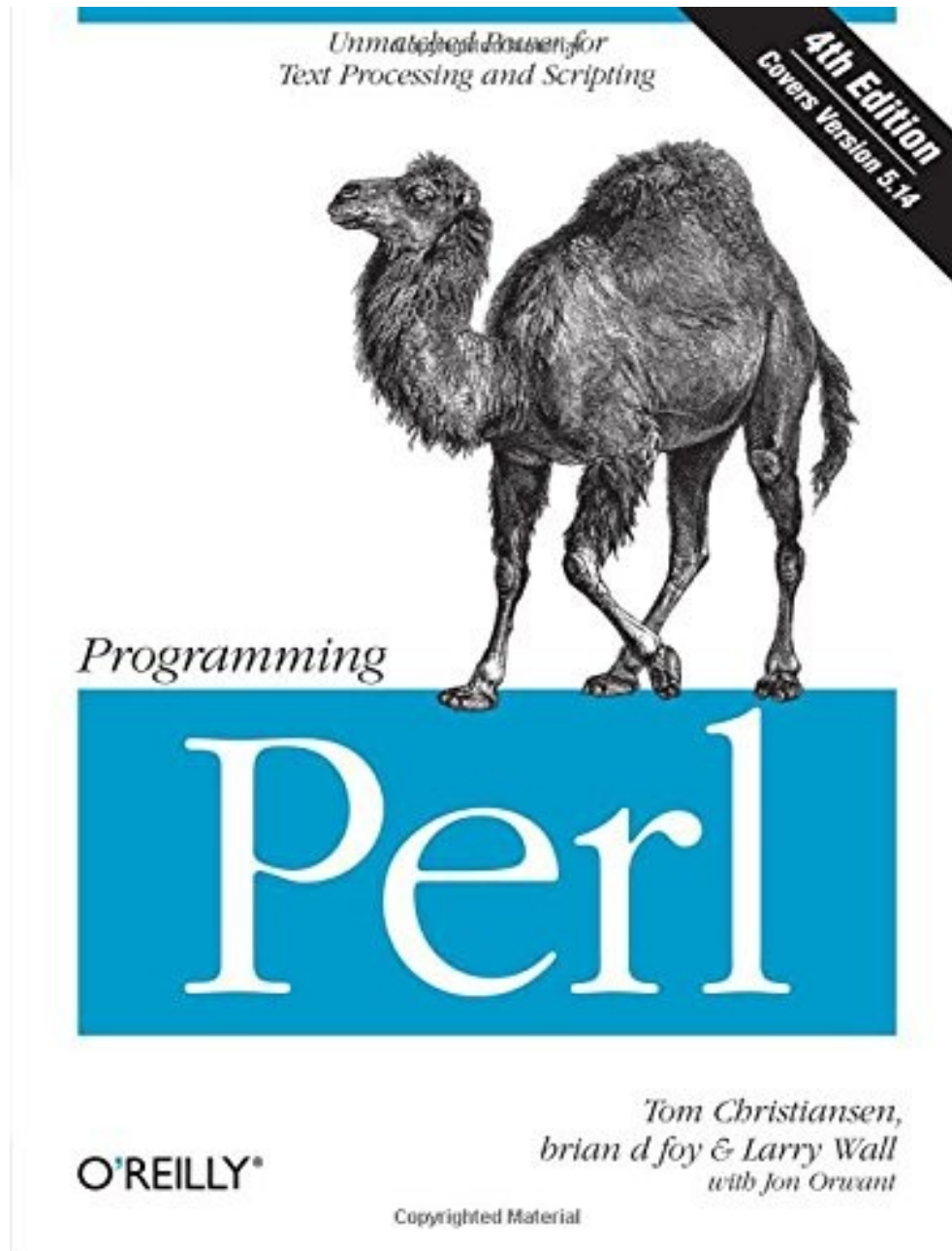
Introducing the IBM
Personal System/2



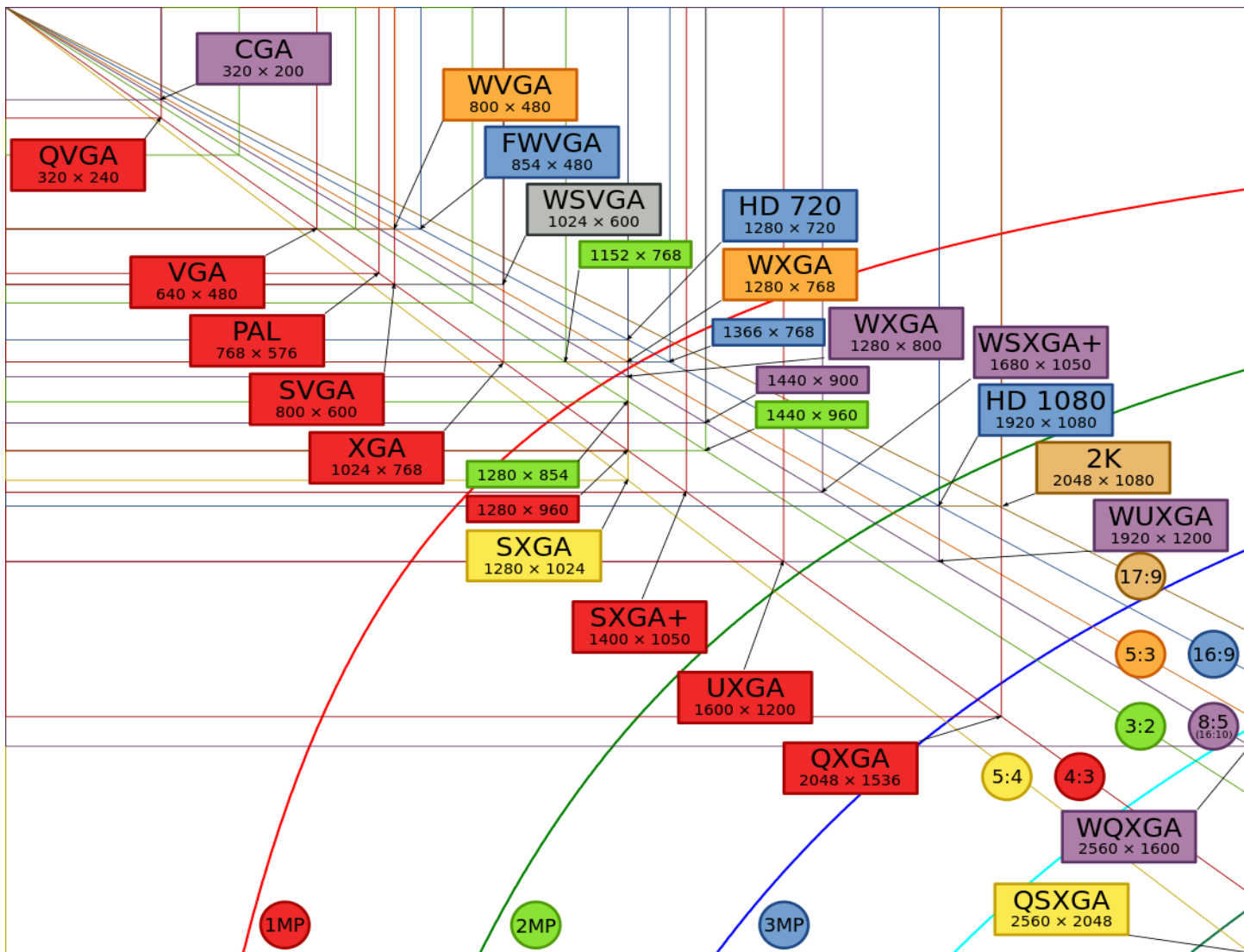
9.49 Perl – 1987



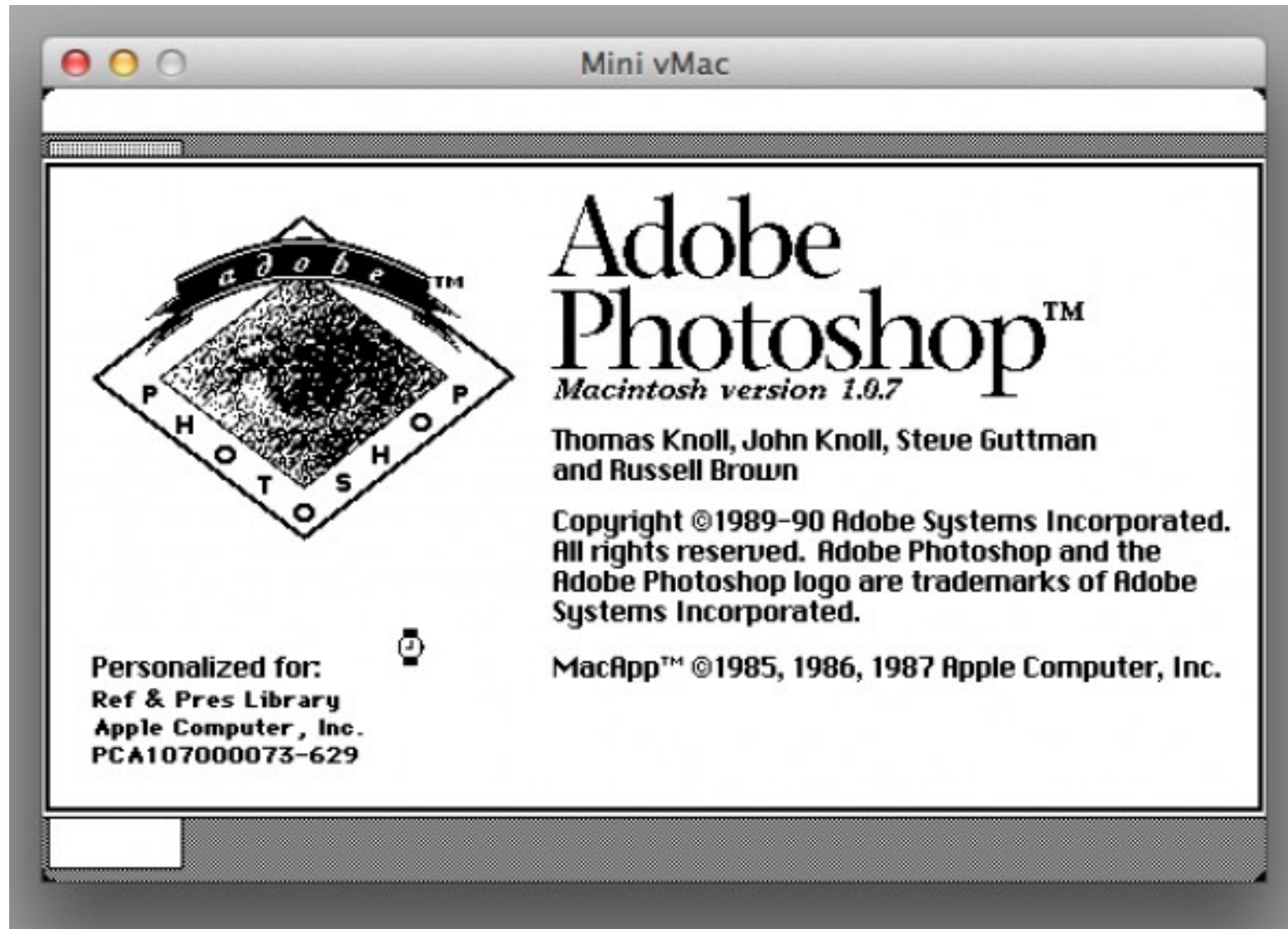
Larry Wall (Estados Unidos, 1954)



9.50 VGA – 1987



9.51 Photoshop – 1987



Thomas Knoll (Estados Unidos, 1960)

9.52 Hypercard – 1987

Hypertext systems
Classifying hypertext systems (Frank Halasz)

Frank Halasz from MCC gave the last talk at the workshop. He and the organizing committee should be criticized for not making it the first talk AND the last talk: Part of the talk was a very good survey of what HT really is and a classification of current systems. This material could have filled a whole talk with no problems but was presented with such speed that it left the audience breathless. It would also have made a good platform for the discussions during the conference if it had been presented at the beginning instead of at the end.

Definition of HyperText

HT systems can be divided into on the one hand the "original" generation of Memex [Vanavar Bush], NLS/Augment* [Engelbart], Xanadu [Ted Nelson], etc. and on the other hand the "current" generation consisting of e.g.

- Research systems: Intermedia [Brown University], NoteCards [Xerox]
- PC Products: Guide [Owl], HyperCard [Apple]
- Workstation products: Document Examiner [Symbolics].

Time since you were here: NEVER

Current report overview map

```
graph TD
    People --> HT[HyperTEXT '87 Workshop]
    Literature --> HT
    TheWorkshop[The workshop] --> HT
    Systems --> HT
    Applications --> HT
    ResearchIssues[Research issues] --> HT
    Definition --> HT
    CSCW[CSCW '86 Trip Report] --> HT
```

Current chapter overview map

```
graph TD
    HT[HyperText systems] --> HTSystems[Classification of HT systems]
    HT --> NoteCards
    HT --> DocEx[Document Examiner]
    HT --> HyperCard[HyperCard Apple]
```

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Front cover
History list

Quit



Steve Jobs

Bill Atkinson

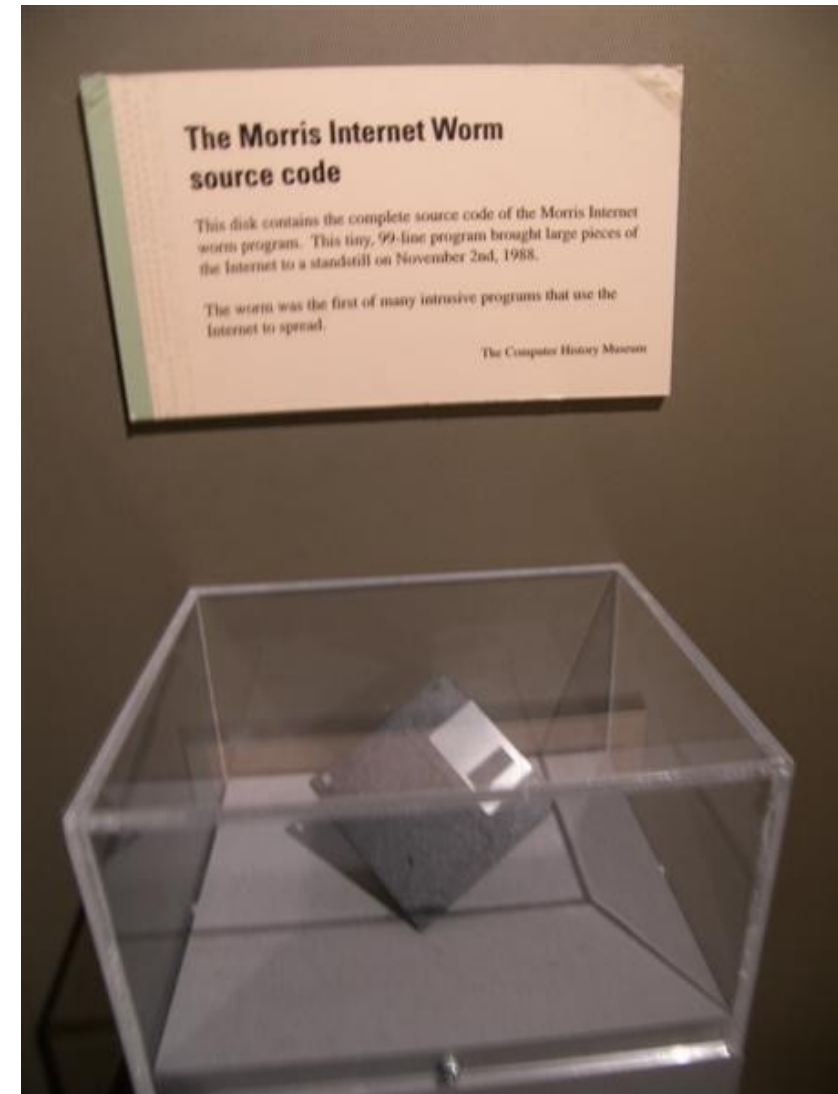
9.53 Worm – 1988

The First Worm?



Robert T. Morris

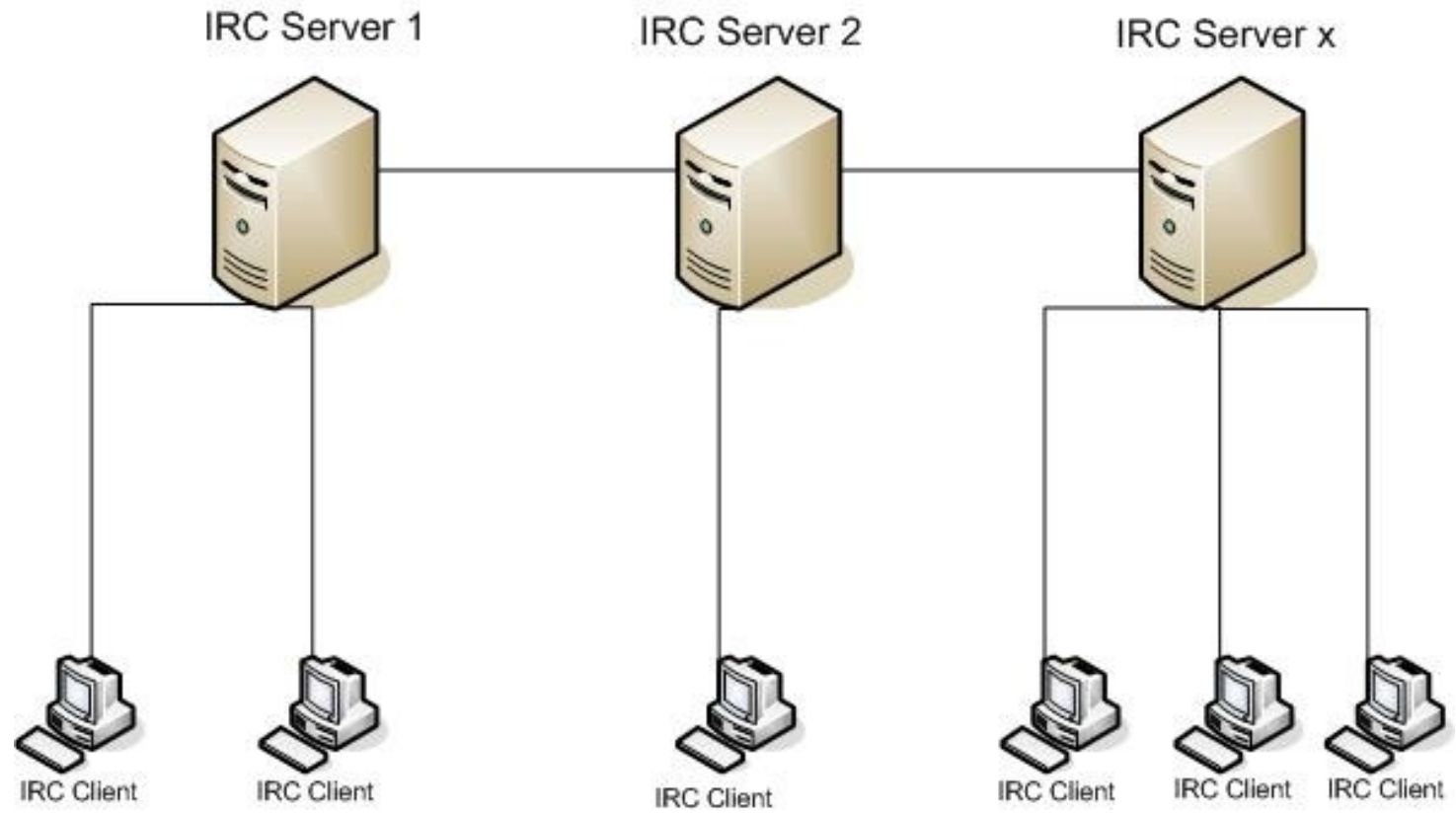
- Morris Worm
 - Written by Cornell student Robert Tappan Morris in 1988
 - Activated the worm from a computer at MIT
 - Found vulnerabilities in Unix and corrupted thousands of computers the first day released
 - Convicted under the 1986 Computer Fraud and Abuse Act, sentenced to community service
 - Present role: Professor at MIT



9.54 IRC – 1988



Jarkko Oikarinen (Finlândia, 1967)



9.55 Archie – 1989



Alan Emtage (Barbados, 1964)



Welcome to archie.icm.edu.pl

Archie Query Form 

Search for:

Database: ☒ Worldwide Anonymous FTP ☐ Polish Web Index

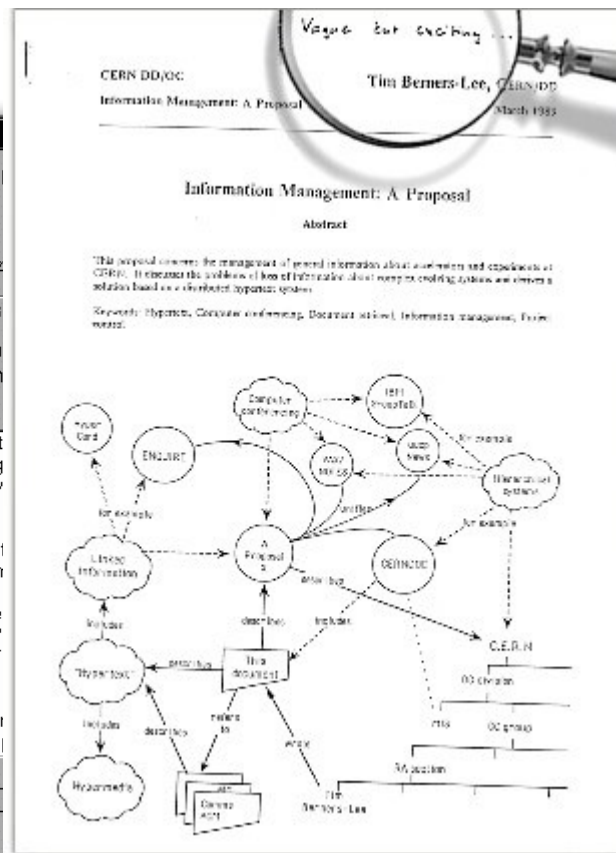
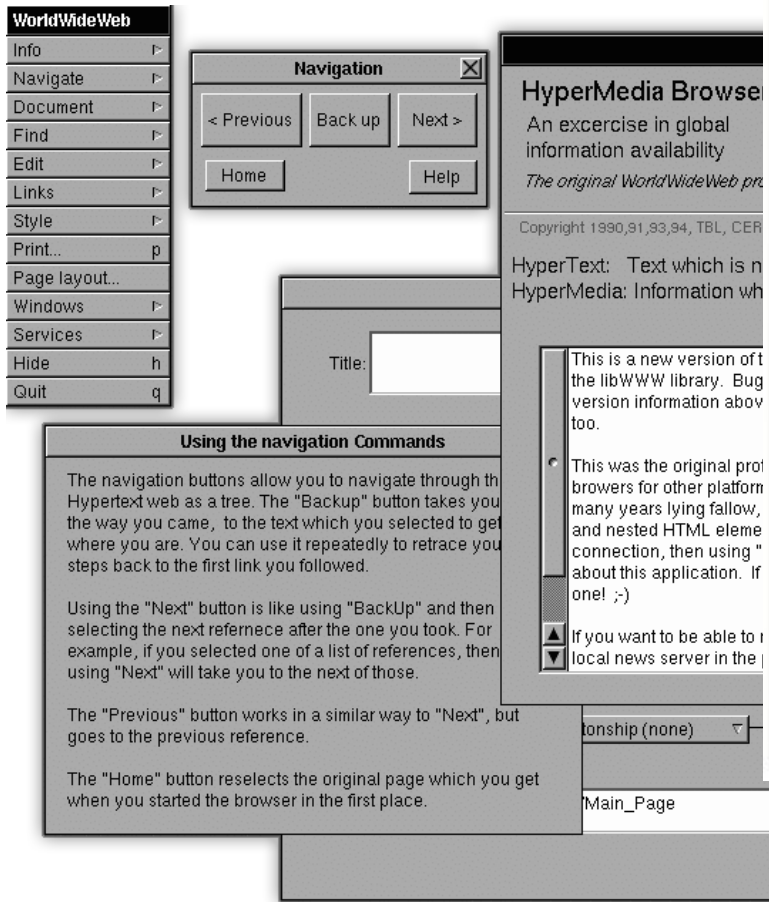
Search Type: ☒ Sub String ☐ Exact ☐ Regular Expression

Case: ☒ Insensitive ☐ Sensitive

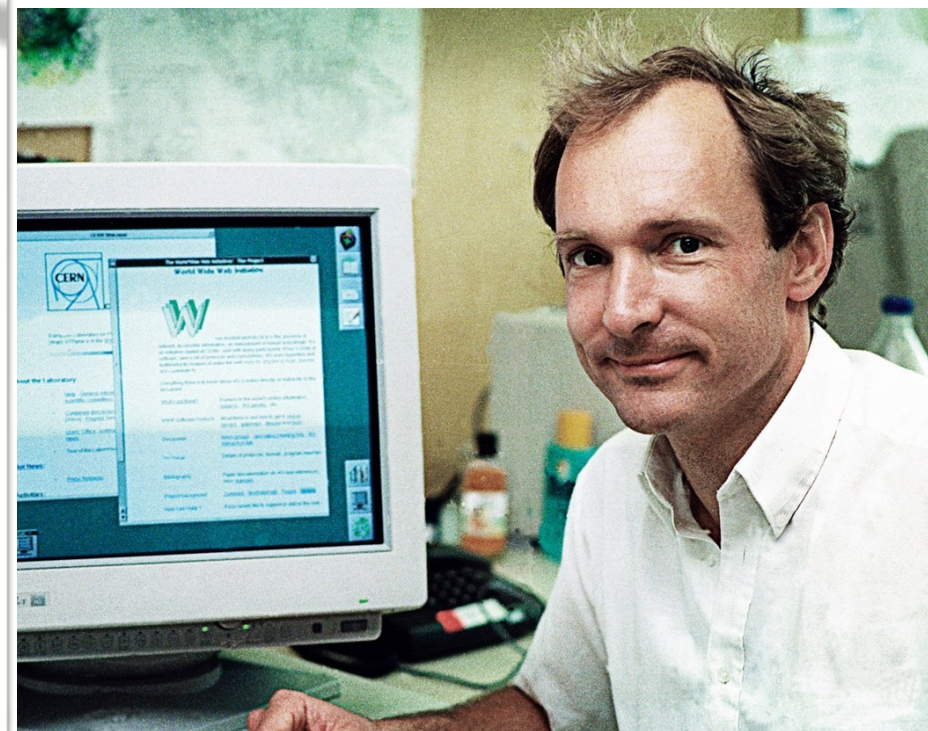
Do you want to look up strings only (no sites returned):
☒ NO ☐ YES

Output Format For Web Index Search: ☐ Keywords Only ☒ Excerpts Only ☐ Links Only

9.57 World Wide Web – 1989



Tim Berners-Lee (Reino Unido, 1955)



9.58 Até aqui...

- Os anos 1980 viram um fantástico desenvolvimento da computação pessoal. O IBM PC foi lançado e seus clones vieram logo depois. A Microsoft cresceu muito com as vendas do DOS e posteriormente do Windows.
- Esse período viu o crescimento das interfaces gráficas. O ícone sendo o Macintosh, embora outros fabricantes também tenham produzido interfaces semelhantes.
- Usuários ganharam poder para produzir aplicações com dBASE II. Outros aplicativos populares como o Word e o Photoshop surgiram neste período.
- Em termos de linguagens de programação, os anos 1980 viram o surgimento de Ada, C++, Turbo Pascal e Eiffel.
- O hipertexto foi popularizado pelo Hypercard, mas realmente tomou o controle das mentes com o World Wide Web. Essa tecnologia estava apenas iniciando no final dos anos 1980, mas mudaria o mundo na década seguinte.